Rolls-Royce

COMPONENT-SPECIFIC RATIONALISED QUALITY STANDARD

Title: Linear Friction Welded Fan Blisk		**RQSC01T-4010**
Part Classification: Critical	Date: 20th June 2025	Issue: 3
Originator: Martin Hibberson	Area: Rotatives, UK	
Engine(s): RB3043 (Pearl 700, Advance 2), RB3070 (Pearl 10X)		CAP No. BRR 247 (disc)

Associated Documents:	Approval	Name	Signature	Date
RQSG000 RQST2T RRP58000 RRP51001 RRP58007 RRP58003	Materials Engineering	T. Wobst	T. Wobst Materials Engineering	Digitally signed by Wobst, Tanja Date: 2025.06.23 10:54:45 +02'00'
MSRR 9987, 9951 See Chapter 9 definitions	Design*	T. Burza	Burza, Tomasz Andrzej	Digitally signed by Burza, Tomasz Andrzej Date: 2025.07.08 09:46:10 +02'00'
Material Specifications: RRMS3600/1 (TDK) MSRR8624 (TMU)	Stress* **(if required by Design, if not signify as NA)**	T. Kaehler	Kaehler, Thomas	Digitally signed by Kaehler, Thomas Date: 2025.07.01 08:39:07 +02'00'
	*Design / Stress signature not required if their approval is recorded on the associated justification report			
	Manufacturing Engineering Approved Delegate	M. Webster	Webster, Matt	Digitally signed by: Webster, Matt DN: CN = Webster, Matt OU = EMEA, UK, Derby, User Accounts Date: 2025.07.09 11:21:23 Z
		Design / Stress approved justification report ~~YES~~ / NO*		
		Service Engineering INFORM ONLY* *Only Required If a change to visual appearance as a result of this issue (delete as applicable)		~~Yes~~ / No

INSPECTION REQUIREMENTS
Indications in any area of the component which are not mentioned in this standard, or when an accumulation of indications raises doubts as to the general standard of components they shall be referred to the Rolls-Royce Technical Authority for assessment and disposition.

This document is subject to the following Export Control Information:

Country	Export Classification	Date
UK	9E003.a.6	20 June 2025
USA	9E991 (NLR)	20 June 2025

1.0 CONTENTS

2.0 PURPOSE

2.1 To define the quality standard for material condition and the associated manufacturing controls necessary to ensure that the Engineering Specification is achieved.

3.0 SCOPE

3.1 This standard must be applied to all materials / components where called up from the Component Definition.

3.2 Work to the requirements of RQST 2T except where indicated in this RQSC.

3.3 This standard is for production components and reflects the expected process capability.

3.4 Components failing to attain the required standard shall be referred to the Rolls-Royce Technical Authority for review, and may be submitted via the non-conformance procedure.

3.5 Any rework of components manufactured prior to issue 3 will have used the zone diagram in section 11.2 of this standard.

4.0 INSPECTION REQUIREMENTS

All inspections (Ultrasonic, Etch, Fluorescent Penetrant, Visual & Binocular) of the components will be in accordance with the relevant Quality Control Test Procedure (QCTP).

4.1 Rectilinear Stage

All forgings shall be inspected by the Forging Supplier to the following recommended sequence, after thermal processing and machining are completed:

a) Surface Condition Assessment to RRP58007.

b) Ultrasonic Inspection to RRP58001.

c) Etch Inspection to RRP51001.

d) Fluorescent Penetrant Inspection to RRP58003.

e) Visual and Binocular Inspection to RRP58007.

4.1.1 Surface Condition (Rectilinear Stage)

Inspection shall be in accordance with RRES90036, RRP 58007 and the relevant QCTP.

Criteria	Accept	Reject
a) Surface finish:	- Drawing requirements met.	- Drawing requirements not met.

4.1.2 Ultrasonic Inspection (Rectilinear Stage)

Inspection shall be in accordance with RRP 58001 and the relevant QCTP. The relevant QCTP shall dictate the setting of the gate. The aim of the inspection is to reject all defects having a response equal to or greater than 0,444mm (0.175”) SAD.

Criteria	Accept	Reject
a) Ultrasonic inspection:	- Standard met.	- Standard not met.

4.1.3 Etch Inspection (Rectilinear Stage)

Process and inspection in accordance with RRP 51001, RRP 58007 and the relevant QCTP.

Criteria	Accept	Reject
a) The etch process if:	- The whole surface is adequately and evenly etched.	- Any sign of incomplete or ineffective etch. - Over-etching (etched surface too dark for evaluation).
b) Macrostructure generally:	- Beta 'ghost' grains as typified by **Figures 1**, **2**, and up to the grain size in **Figure 3**. - 'Tree Ringing' as per **Figure 5**. - Components that are shown to be clear of defects.	- Beta 'ghost' grains larger than those shown in **Figure 3**. See coarser grain size shown in **Figure 4.** - 'Tree Ringing' as per **Figure 6**. - Components showing evidence of overheating as per **Figure 7**. - Areas of coarse grain where there is a sharply defined boundary between fine and coarse grain areas i.e. non- uniform grain size with isolated individual or groups of coarse grain as per **Figure 8**. - Beta fleck as per **Figure 9**. - High Interstitial Defect (HID) as per **Figures 10** and **11**. - All other observed defects e.g. cracks, voids and porosity.

4.1.4 Penetrant Inspection (Rectilinear Stage)

Process and inspection in accordance with relevant QCTP and RRP 58003.

Criteria	Accept	Reject
a) The penetrant process, if the whole surface being inspected:	- Has a uniform surface colour. - Is relatively clean. - Has a residual background colour of the penetrant i.e. slightly fluorescent not removed by washing process, but against which the surface flaws can be readily seen	- Has a high background of fluorescent colour/condition.
b) General:	- Components clear of indications.	- All defects seen.

4.2 Disc Stub Machined Prior to Linear Friction Weld (LFW)

4.2.1 Refer to Section 2 of RQST2T for inspection of the LFW faces prior to weld.

4.3 Linear Friction Weld

4.3.1 Refer to Section 2 of RQST2T for Linear Friction Welding requirements.

4.4 Heat Treatment Cycle

4.4.1 Heat Treatment Cycle Abort

4.4.1.1 ACCEPT – One heat treatment cycle abort, providing that this occurs during heat up and the component temperature is at or below 400^{o}C. This aborted cycle must be prior to completing a heat treatment cycle that conforms to data card parameters.

4.4.1.2 REJECT – Heat treatment cycles that are aborted but do not comply 4.4.1.1

4.4.2 Load Thermocouple Failure

4.4.2.1 The failure of a single load thermocouple can be accepted providing the following conditions are met:

4.4.2.1.1. There were two or more load thermocouples within the cycle, and therefore at least one still remained.

4.4.2.1.2. Rolls-Royce Laboratory or the Rolls-Royce Controlling Manufacturing Engineer (M.E.) has assessed the effect that losing this load thermocouple has had on the cycle when comparing it to data card parameters. This should take into account whether the load thermocouple controls time at temperature, and whether the remaining cycle data appears similar to a previous acceptable cycle with no thermocouple failures:

4.4.2.1.3. ACCEPT if no effect is seen on the heat treatment cycle from the load thermocouple failure.

4.4.2.2 REJECT if this load thermocouple failure may have had an effect on the heat treatment cycle.

4.4.3 Partial loss of Vacuum

4.4.3.1 ACCEPT – Partial loss of vacuum if this does not result in any discolouration or if any discolouration seen is acceptable to the image given in **Figure 12**.

4.4.3.2 REJECT - Heat treatment cycles that have lost some vacuum, but do not comply 4.4.3.1.

4.4.4 Post Heat Treatment Cycle Assessment

4.4.4.1 ACCEPT – Heat treatment cycle that either conforms to all data card parameters or conforms to all data card parameters, apart from a heat treatment cycle abort, load thermocouple failure or partial loss of vacuum but is still acceptable to sections 4.4.1, 4.4.2, and 4.4.3 in this standard.

4.4.4.2 REJECT – Heat treatment cycles that do not comply with 4.4.4.1

4.5 Post Heat Treatment Visual Inspection

4.5.1 Post Heat treatment Visual Inspection Requirements

4.5.1.1 Removal of the discolouration is acceptable by the Approved Fixed Process, post heat treatment processing.

4.5.1.2 ACCEPT:

4.5.1.2.1. Silver colour. Example shown **Figure 12.**

4.5.1.2.2. Shades of straw, yellow, gold, blue and purple discolouration up to the images given in **Figures 12 and 13**, providing the following conditions have been met:

Inconsistent discolouration, for example in **Figure 12**, in which some of the blades are discoloured (gold), some which are not (silver) can be caused by difference in surface finish. Providing the heat treatment has been carried out in accordance with the approved heat treatment control procedures, this can also be accepted.

4.5.1.3 REJECT: Any other discolouration outside of the acceptance criteria in 4.5.1.2.

4.6 Finish Machined Stage

4.6.1 Surface Condition

4.6.1.1 Refer to RQST 2T section 2.4.6, except where there are any references to JES137, work to the requirements of RRES90036 instead of JES137. For any component staining assess to RQST 2T section 2.4.8. Also see section 4.5 of this standard for post heat treatment visual inspection.

4.6.1.2 Assessment of features with a stylus probe of 1,0mm diameter or less, using the technique described in RQST 2T section 2.2 is to be used. NOTE: This technique is used to quickly determine how deep a feature is. If there is any confusion regarding if a stylus probe is hesitating, then the ultimate determination is how deep the feature is.

4.6.1.3 ACCEPT - Scratches, scores, nicks and indents on all peened surfaces, both prior to and post peen that do not cause the stylus probe to hesitate or, if the probe does hesitate, are less than 0,025mm deep. See images in **Figure 19 and 20** for typical examples on aerofoils.

4.6.1.4 ACCEPT - Scratches, scores, nicks and indents on all non-peened surfaces that do not cause the stylus probe to hesitate or, if the probe does hesitate, are less than 0,013mm deep. NOTE: This acceptance allowance is for all non-peened surfaces.

4.6.1.5 ACCEPT – The visual appearance of any dimensional changes, such as mismatches or undulations. The dimensional conformity of the component is determined by dimensional inspection. Examples of some features are given in **Figure 22**.

4.6.1.6 ACCEPT – Machining marks on blisk, providing surface finish requirements are met.

4.6.1.7 ACCEPT - Machining marks on all aerofoil surfaces if they do not cause the stylus probe to hesitate.

4.6.1.8 ACCEPT – Machining marks on tip milled aerofoils are visually acceptable. An example of an acceptable blade tip is given in **Figure 14.**

4.6.1.9 ACCEPT – Machining chatter in all holes, up to the level given in RQST 2T Figure 68, except the holes in the curvic coupling and damper flange holes (where a damper flange is present) where no visible chatter is acceptable. Note: Figure 69 of RQST 2T is still unacceptable for all holes.

4.6.1.10 ACCEPT – Witness of machining marks on polished aerofoil leading and trailing edges, and dwell marks in the fillet and annulus, provided they are smooth and continuous, and do not snag a stylus, see examples in **Figure 23**.

4.6.1.11 ACCEPT - Flats and other visual anomalies on aerofoil trailing edges providing that they are in line with JES 195. See examples in **Figure 24** of acceptable conditions.

4.6.1.12 ACCEPT – Marks from deburring of curvic teeth. See image in **Figure 25** of acceptable curvic teeth with marks from deburring.

4.6.1.13 ACCEPT – Machining marks on the aerofoils, see images of typical marks given in **Figure 21**. These marks are acceptable at final inspection up to 15 microns deep.

Note: These machining marks are from blade manufacture and must not to be assessed using a stylus probe. Also, for in process inspections prior to shot peening these can be up to 25 microns deep.

4.6.1.14 ACCEPT – Leading edge and trailing edge bolt holes with score marks up to 65 microns deep providing these were only found after the shank nuts or balance bolts were removed.

4.6.1.15 REJECT – Any other defects seen.

4.6.2 Etch Inspection

4.6.2.1 Refer to RQST2T Section 2.4.2.

4.6.2.2 See Section 4.6.1

4.6.2.3 ACCEPT – Beta ‘ghost’ grains to sizes given in RQST2T Figures 1, 2 and 3.

4.6.2.4 ACCEPT – Machining black spot in all holes except the holes in the curvic coupling, up to the level given in RQST2T Figure 20, providing that these do not hold penetrant.

4.6.2.5 ACCEPT - Areas of additional material, providing that the following criteria are met:

(i) The additional material does not inhibit the inspection taking place e.g. Grain structure can still clearly be assessed.

(ii) The individual items of additional material are spread over the surface being inspected and not clustered together.

(iii) There are no other defects associated with the indication.

(iv) It is not smeared.

(v) It is not associated with work hardening.

Note: An example of ACCEPTABLE additional material is shown in **Figure 15.**

4.6.2.6 ACCEPT - Additional material and black spot in areas that will be subsequently machined, provided that the extent of the additional material and black spot does not inhibit the inspection of that area.

4.6.2.7 REJECT – Any other defects seen which are outside this requirement, e.g. smeared material.

4.6.3 Penetrant Inspection

4.6.3.1 Refer to RQST2T Section 2.4.3

4.6.3.2 See Section 4.6.1

4.6.3.3 ACCEPT – Areas of additional material, providing the following criteria are met:

(i) The additional material does not hold penetrant.

(ii) The additional material holds penetrant, but the penetrant indication is only associated with the geometric features of the additional material, and that no other defects are associated with the indication. Use magnification to determine this, when required.

(iii) The individual items of additional material are spread over the surface being inspected and not clustered together.

(iv) The additional material penetrant indications do not inhibit the inspection taking place, e.g. There are a small number of features to assess, and these are not affecting the ability to inspect the remainder of the component.

4.6.3.4 REJECT – Any other defects seen which are outside this requirement, e.g. smeared material or machining black spot.

4.6.4 Shot Peen

4.6.4.1 Refer to RQST2T Section 2.4.6 for surface condition pre-shot peen.

4.6.4.2 Refer to RQST2T Section 2.4.6 for surface condition post shot peen

4.6.5 Transition (Turbulator) Strips

4.6.5.1 This section applies when the finish drawing definition requires transition (turbulator) strips to be applied to aerofoils. Each aerofoil and transition strip will be visually inspected by the strip supplier to an agreed inspection procedure, e.g. on a data card. **Figure 26** gives images of typically acceptable transition strips. The component shall be inspected to finish drawing definition, and to the following requirements:

4.6.5.2 ACCEPT

(i) Full coverage of specified area.

(ii) Smoke discolouration up to 4mm around the strip.

(iii) Local raised edges of the strip surface.

(iv) Dressed surface of the strip, resulting in a “polished” appearance.

4.6.5.3 REJECT

(i) Cracking

(ii) Chipping

(iii) Blistering

(iv) Porosity

(v) Lack of adhesion

(vi) Overspray on adjacent surfaces.

(vii) Any other observed defect outside the ACCEPT criteria in 4.6.5.2

5.0 SALVAGE / RECTIFICATION

5.1 Refer to RQST2T Section 3.

5.2 In addition, the permitted removal of local damage and surface discontinuities (see RQST2T, section 3.2.(a)) does not include cracks.

5.3 In addition to the permitted rectification at the finished machined welded assembly stage given in Section 3 of RQST2T the following are allowed:

5.3.1 After rectification the component must still comply with Engineering drawing requirements. However, where from existing data it is known that the surface finish and / or dimensions will still be within drawing, it is not necessary to re-measure with agreement of Rolls-Royce Manufacturing Engineering.

5.3.2 In areas with stock-on condition no rework is required if the depth of the damages are less than 50% of the stock. Nevertheless it has to be ensured that the required NDT operations can be performed. If this is not possible due to the imperfections, they have to be dressed to single depth.

5.3.3 CHATTER – Chatter of up to 0,013mm peak to trough may be removed by light hand polishing or butterfly polishing using approved abrasive products. If no power tools are used confirmation of its removal is by a visual inspection at x10 magnification minimum, no further NDT is required.

5.3.4 FEATURE ASSESSMENT – The use of scotch-brite is permitted in order to aid feature assessment, as well as remove superficial surface marks, blemishes and to improve cosmetic appearance.

5.3.5 ADDITIONAL MATERIAL – It is acceptable to polish to remove additional material found at post (descale and) combination etch binocular inspection without repeating this inspection providing that the requirements of section 4.6.2 have been met. Polishing must be carried out by hand polishing (ie. no power tools) using approved abrasive products. However, this polishing must be carried out prior to preparation for penetrant inspection.

Note: After penetrant inspection has been completed it is also acceptable to polish to remove any additional material found at penetrant inspection without repeating this inspection providing that the requirements of section 4.6.3 have been met. Polishing must be carried out by hand polishing (ie. no power tools) using approved abrasive products.

5.3.6 SHOT PEENED AREAS– Scratches scores, nicks and indents on peened surfaces that have been reworked up to a depth of 0,025mm on aerofoil surfaces between the top of the fillet and the blade tip (feature type AA) do not need to be re-peened, see **Section 11**. Also, those in any low stressed zones, to a depth up to 0,100mm will not require re-peening, see **Section 11**. For all other areas, and depths above these requirements, refer to RQST2T section 3.

5.3.7 Is it acceptable to repeat any machining operations, including robot polishing providing it is prior to etching, non-destructive testing and shot peen, and the machining parameters used are already validated for this component. However, it would not be acceptable to repeat machining operations on peened surfaces after shot peen.

5.3.8 In instances where the fixed process polishing operation to RRP 55003 post shot peen has left areas of peen visible. Hand dressing without using power tools is allowed to improve the surface finish. Provided a consistent texture is produced and drawing surface finish requirements are met. This is in addition to the hand dressing that is already allowed to the Fixed Process Document (FPD).

5.3.9 Where the rectification allows removal of high spots only, it is acceptable to confirm the depth of scratches, scores, nicks and indents after the removal of the high spots only.

5.4 Finished machined condition dressing allowances

Finished machined dressing allowances are given below, see **Section 11**. Re-work and inspection requirements for surface anomalies, i.e. scratches, scores, nick and indents are given in **Figure 16 and Section 11**.

5.4.1 Feature Type AA – Aerofoil, and fillet radius to aerofoil.

5.4.1.1 Dressing is allowed to a maximum depth of 0,13mm. Dressing must be a smooth blend with the surrounding profile.

5.4.1.2 In addition to the allowances given in **Figure 16** it is permissible to dress up to two areas at up to 0,060mm deep on each aerofoil between the top of the fillet and the blade tip and not carry out any further NDT providing that a visual inspection at x10 magnification minimum is carried out it to ensure that artefact has been removed.

5.4.2 Feature Type BB - Holes and adjacent area which is concentric to the holes with a diameter of 5mm plus the diameter of the hole.

5.4.2.1 Dressing of holes is allowed, including to remove blackspot, in accordance with RQST2T sections 3.4(i), 3.4.2.3. and 3.7.

5.4.2.2 Any dressing in the bore of the hole must be blended to the full depth and circumference of the hole.

No dressing is allowed in holes which are in the shot peened condition.

5.4.3 Feature Type DD- Diaphragm and hub areas

Note: This is for zone diagrams in Section 11.2 only

5.4.3.1 Dressing is permitted to a maximum depth of 0,25mm.

5.4.3.2 The extent of dressing shall be as near as possible to a circular area of a diameter of not less than 100 times the depth of dressing, although it is not necessary to record the actual length to depth ratio.

5.4.3.3 Only one area of dressing is permitted in the feature.

5.4.4 Feature Type EE – Drive Arm

Note: This is for zone diagrams in Section 11.2 only

5.4.4.1 The depth of dressing must not exceed 5% of the drawing maximum section thickness or 0,13mm whichever is the smaller.

5.4.4.2 The extent of dressing shall be as near as possible to a circular area of a diameter not less 100 times the depth of dressing, although it is not necessary to record the actual length to depth ratio.

5.4.4.3 The amount of salvage is limited to two areas of dressing. This may be two areas per face or one area on each face provided they are not axially opposite.

5.4.5 Feature Type LL – Platform (annulus)

5.4.5.1 Dressing is permitted up to a maximum depth of 0,13mm.

5.4.5.2 The extent of dressing shall be as near as possible to a circular area of a diameter not less than 50 times the depth of dressing, although it is not necessary to record the actual length to depth ratio.

5.4.6 Feature Type MM – Under rim

Note: This is for zone diagrams in Section 11.1 only

5.4.6.1 Dressing is permitted to a maximum depth of 0,13 mm.

5.4.6.2 The extent of dressing shall be as near as possible to a circular area of a diameter not less 100 times the depth of dressing, although it is not necessary to record the actual length to depth ratio.

The amount of salvage is limited to two areas of dressing. This may be two areas per face or one area on each face provided they are not axially opposite.

5.4.7 Feature Type OO – Leading edge fingers and adjacent radii

5.4.7.1 Dressing is permitted to a maximum depth of 0,25mm

5.4.7.2 The extent of dressing shall be as near as possible to a circular area of a diameter not less than 50 times the depth of dressing, although it is not necessary to record the actual length to depth ratio.

5.4.7.3 In the event of both sides of the fingers being dressed such areas must not be axially opposite. This is to prevent thinning from both surfaces.

5.4.8 Feature Type PP – Trailing edge balance flange

5.4.8.1 Dressing is permitted to a maximum depth of 0,25mm.

5.4.8.2 The extent of dressing shall be as near as possible to a circular area of a diameter not less than 50 times the depth of dressing, although it is not necessary to record the actual length to depth ratio.

5.4.8.3 In the event of both sides of the balance flange being dressed such areas must not be axially opposite. This is to prevent thinning from both surfaces.

5.4.9 Feature Type QQ – Curvic Coupling

5.4.9.1 Scratches, scores, nicks and indents up to 0,025mm deep may be dressed on surfaces A, B, C and D (excluding the restricted dressing zone), see **Figures 17 and 18**. Dressing may be to completely remove the feature, or to remove high spots only. Areas dressed to remove high spots only require assessment by x10 magnification visual to ensure that the high spots only have been removed. Areas dressed to completely remove feature shall be assessed by swab etch and binocular inspection.

NOTE: Scratches, scores, nicks and indents up to 0,013mm deep can be accepted as per clause 4.6.1.4.

5.4.9.2 In addition to the allowances given in 5.4.9.1, and for the same surfaces as per 5.4.9.1, the maximum depth of a scratch, score, nick or indent that can be dressed is widened from 0,025mm to 0,080mm. Dressing is to completely remove the feature. Areas dressed shall be assessed by swab etch, binocular and penetrant inspection.

5.4.9.3 In addition to the allowance given in 5.4.9.1-5.4.9.2, and for the same surfaces as per 5.4.9.1, but on a maximum of four curvic teeth per component: The maximum depth of a scratch, score, nick or indent that can be dressed is widened from 0,080mm to 0,180mm. Dressing is to completely remove the feature. Inspection requirements are as defined under 5.4.9.2.

5.4.9.4 In addition to the allowance given in 5.4.9.1-5.4.9.3, on a maximum of four curvic teeth per component, but in the restricted dressing zone of surface D (**Figures 17 and 18):** Scratches, scores, nicks or indents up to 0,025mm deep are to be dressed to remove high spots only. Inspection requirements are as defined under 5.4.9.1 for high spot removal.

5.5 Following any correction using swab etch, any visual witness of etch must be removed by dressing with Scotch-brite.

5.6 Transition (Turbulator) Strips

5.6.1 Transition Strip Rectification

5.6.1.1 Transition strip(s) in the correct location but not acceptable to the requirements of 4.6.5 can be removed electrolytically in accordance with an approved data card. The strip(s) are to be re-applied to the approved data card. One removal and re-application is allowed per blade.

5.6.1.2 Transition strip(s) in the wrong location are to be referred to the Rolls-Royce Technical Authority.

6.0 MANUFACTURING CONTROL

6.1 Refer to Manufacturing Limitations in RQST2T, section 4, with the following additional requirements:

6.1.1 Any special manufacturing controls will be referenced in the component Fixed Process Document (FPD) in accordance with RRES90000.

6.1.2 For additional limitations refer to the relevant component QCTP.

7.0 COMPONENT APPROVAL REQUIREMENTS

7.1 Refer to component approval in RQST2T, section 5, with the following additional requirements:

7.1.1 The component approval requirements are contained within the CoSAPs (Condition of Supply Approval Package) quoted on Page 1 of this RQSC.

7.1.2 The requirements for initial and where applicable subsequent sequential and integrity testing shall be contained within the CoSAPs referenced on Page 1 of this RQSC.

7.1.3 The requirement of MSRR9951, MSRR9987, RRMS30000, RRMS30001 and this specification shall also be met.

8.0 COMPONENT RELEASE REQUIREMENTS

8.1 There are no additional release requirements. All components released shall conform to the relevant Materials specification, this RQSC and the relevant sections of RQST2T. Manufacturing shall carry out sufficient checks to ensure the requirements of the Component Definition and its associated specifications are achieved.

9.0 DEFINITIONS

9.1 For Definitions refer to RQSG000 and RQST2T, section 6

9.2 In addition the following definitions apply:

9.2.1 Non-peened surfaces – These are surfaces that are not and will not be peened as part of the component method of manufacture.

9.2.2 Partial loss of vacuum – This is a worsening of vacuum, but not a complete loss.

9.2.3 Load thermocouple failure – This is either when a load thermocouple fails by giving a reading that is clearly incorrect, or the load thermocouple becomes detached from the component. Examples of these are (i) A furnace is running at 600^{o}C but a load thermocouple is giving a reading of 4000^{o}C, or (ii) A load thermocouple gives a reading which is outside the temperature tolerance as it has become detached from a component and has moved to outside the furnace working volume. Note: These are examples, not a complete list.

10.0 APPENDICES

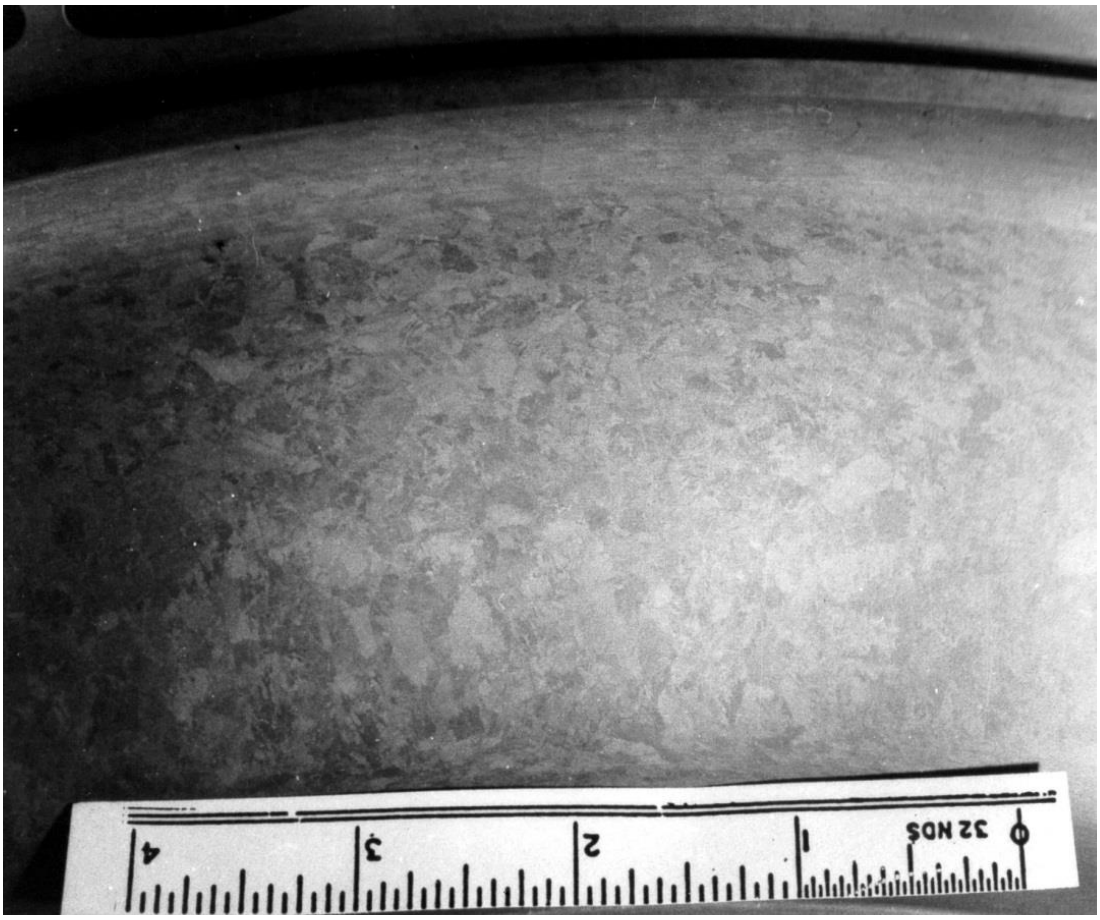

Figure 1: ACCEPTABLE macrostructure showing beta ‘ghost’ grains (Mag. x1).

NEG No: 93.1000.9

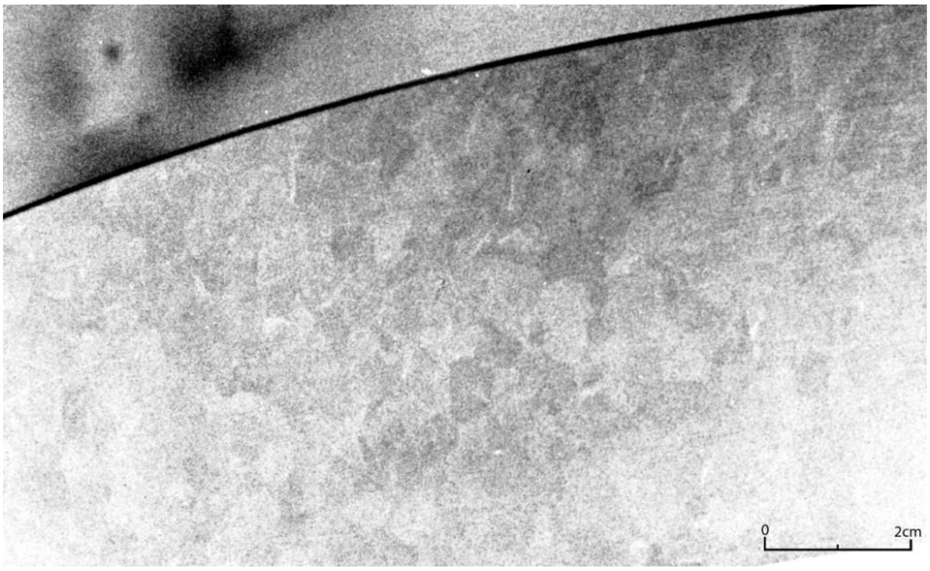

Figure 2: ACCEPTABLE macrostructure showing beta ‘ghost’ grains (Mag. x1).

NEG No: 800.1368

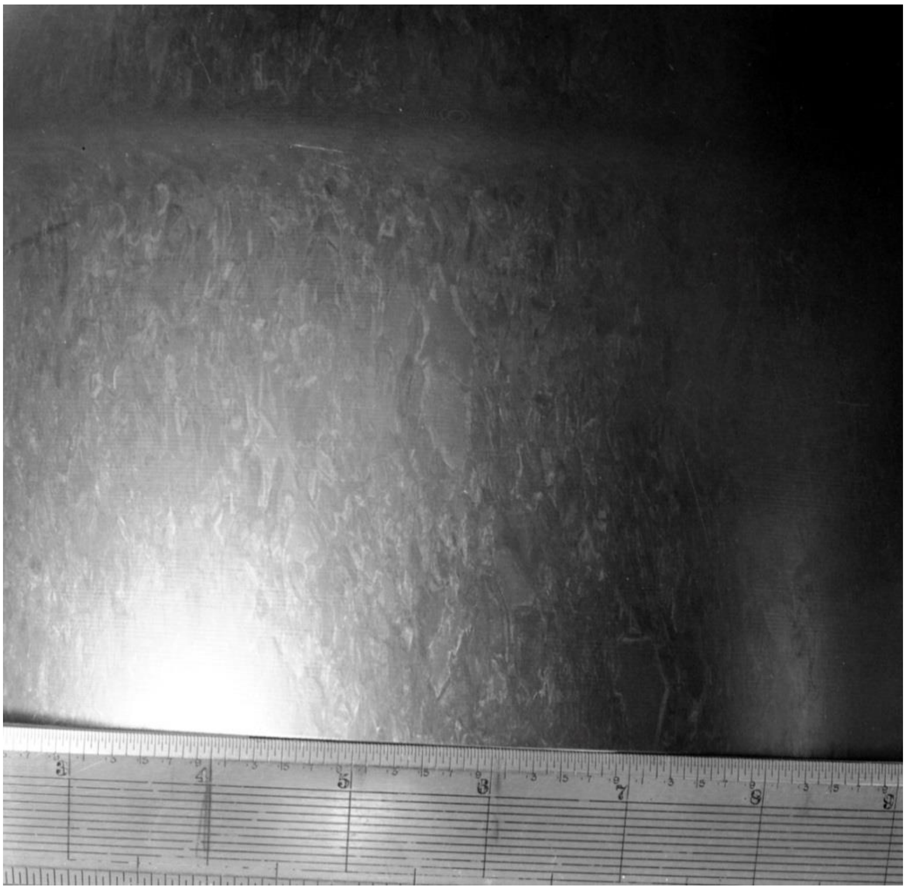

Figure 3: ACCEPTABLE macrostructure showing beta ‘ghost’ grains (Mag. x1).

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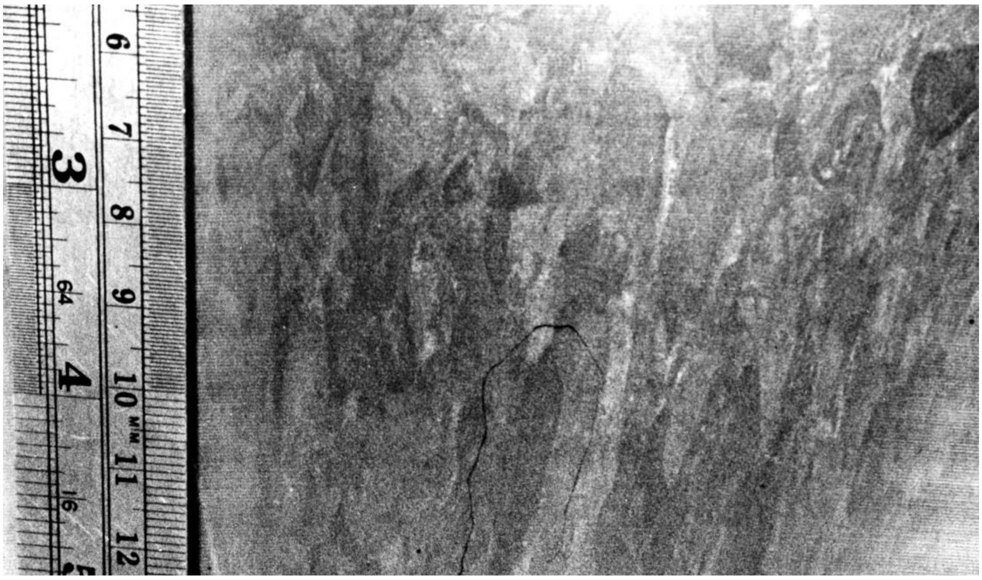
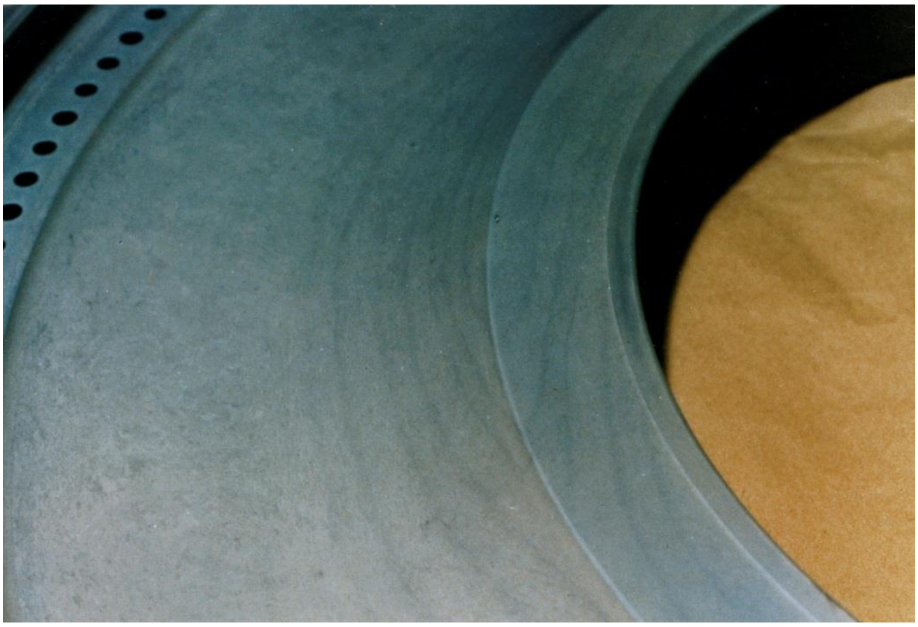


Figure 4: REJECTABLE macrostructure showing beta ‘ghost’ grains.

NEG No: 83.550.8.



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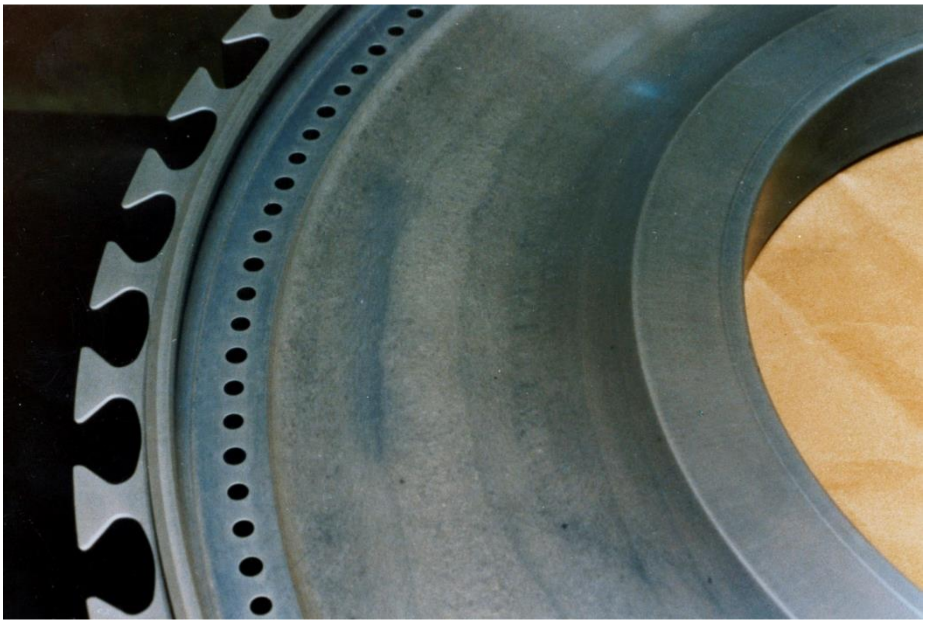


Figure 5: Examples of ACCEPTABLE macrostructure showing tree ringing (Mag. x1) ***continued overleaf***

NEG No: 83.550.6.

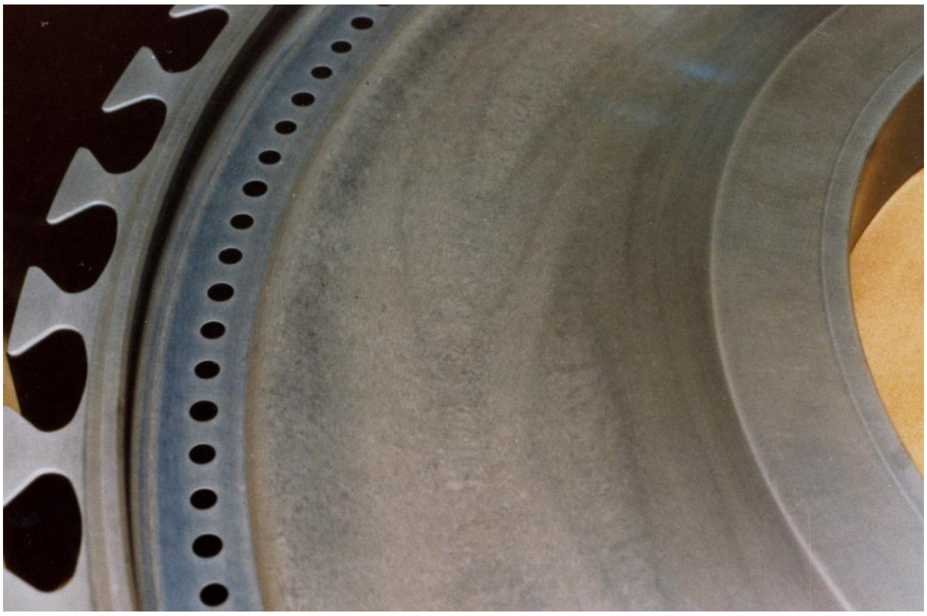
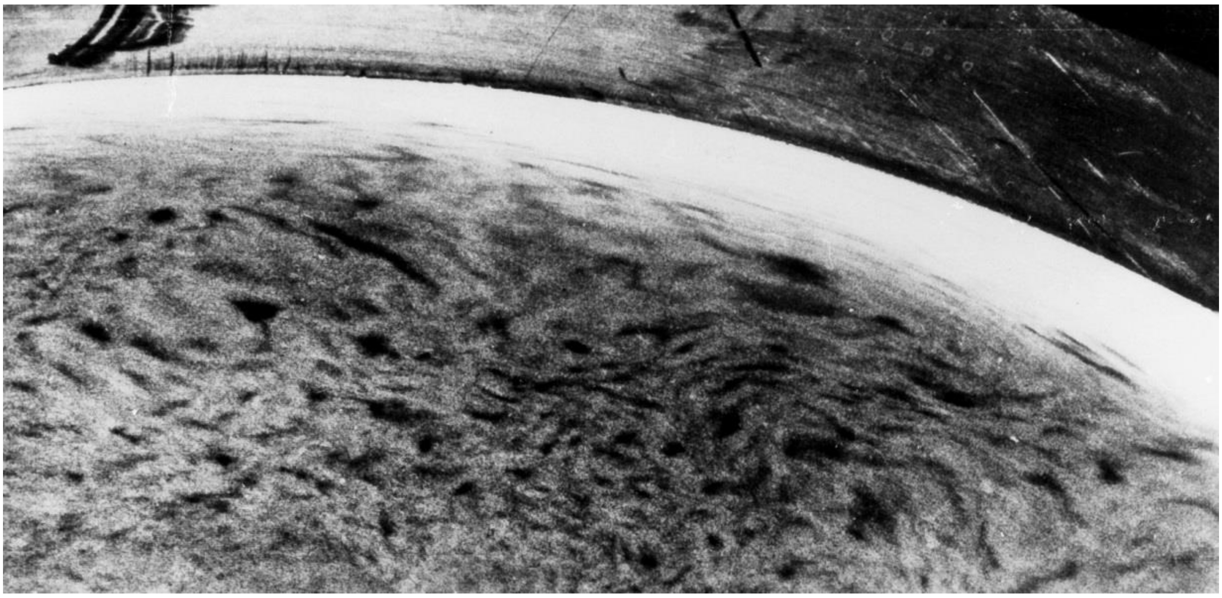


Figure 5: Examples of ACCEPTABLE macrostructure showing tree ringing (Mag. x1).

NEG No: 790.7108. (Copy Neg: 91.0846.11)



NEG No: 790.7108. (Copy Neg: 91.0846.11)

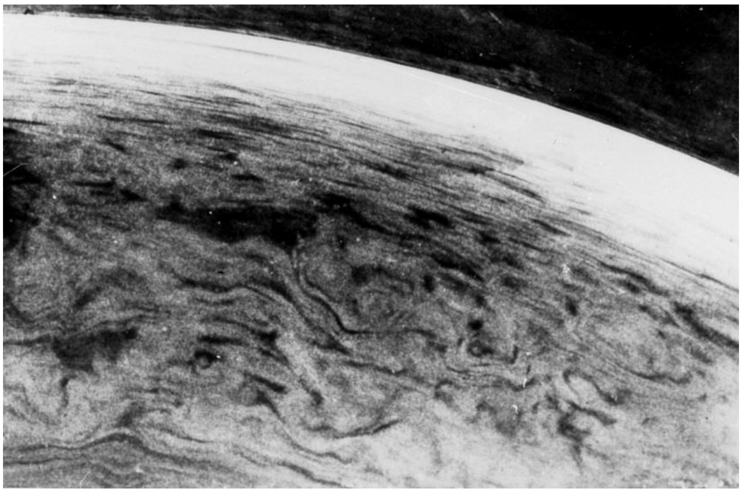


Figure 6: Example of UNACCEPTABLE macrostructure showing tree ringing (Mag. x1).

NEG No: 87.1852.7. (Copy Neg: 91.0846.3)

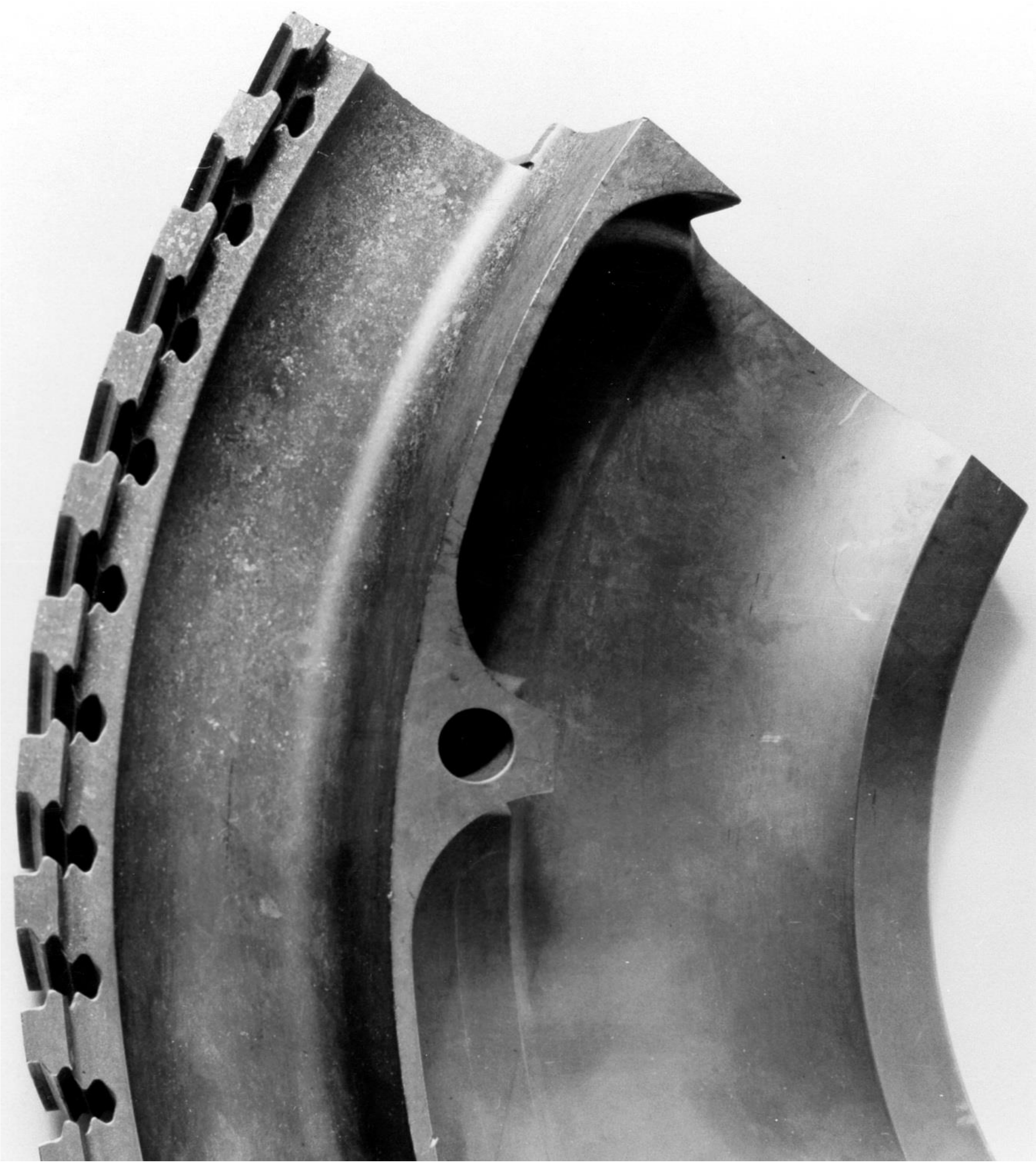


Figure 7: Example of UNACCEPTABLE macrostructure with component showing evidence of overheating.
Continued overleaf

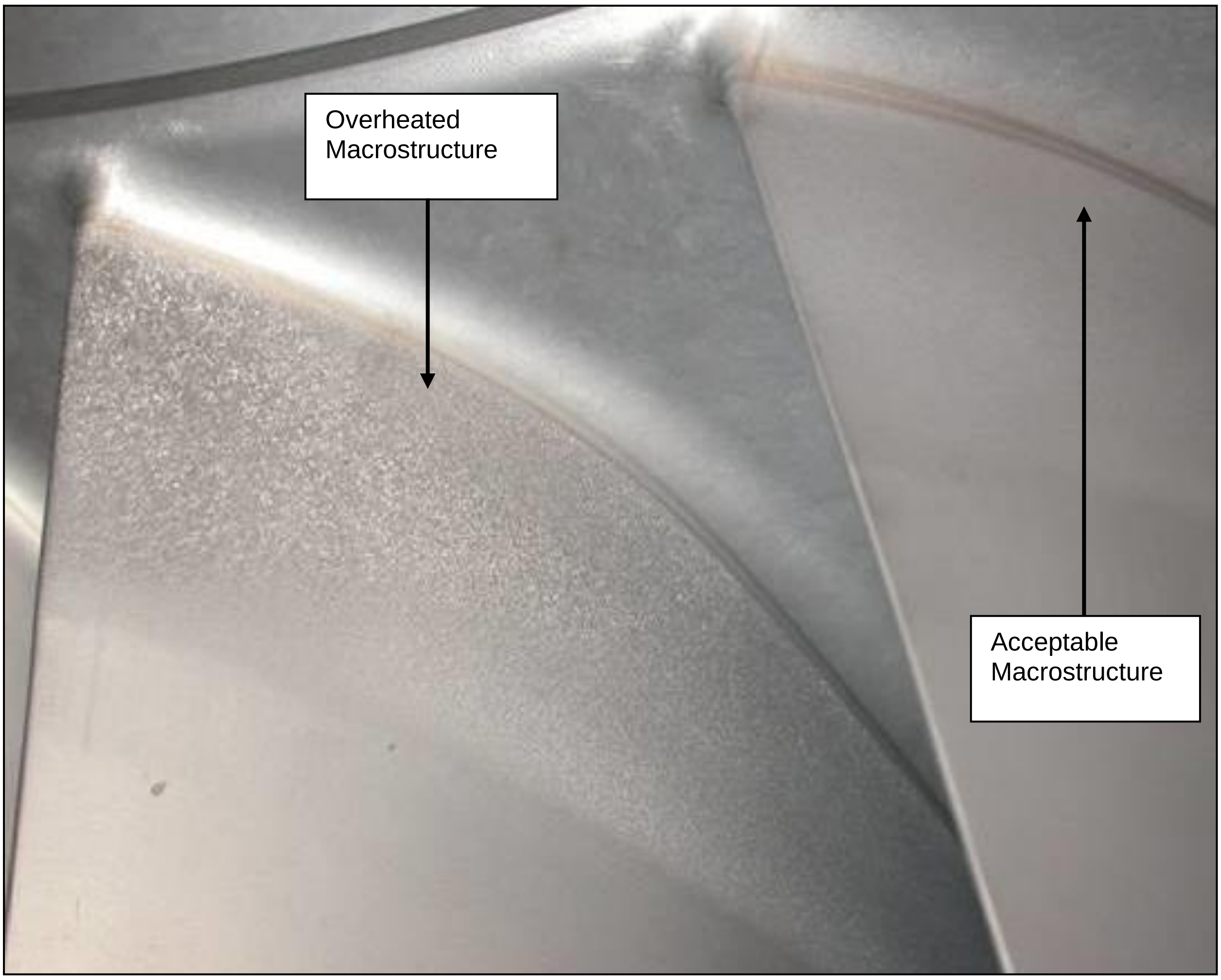


Figure 7: Example of UNACCEPTABLE macrostructure with component showing evidence of overheating. (magnification x2)

NEG No: H 17263. (Copy Neg: 93.0856.2)

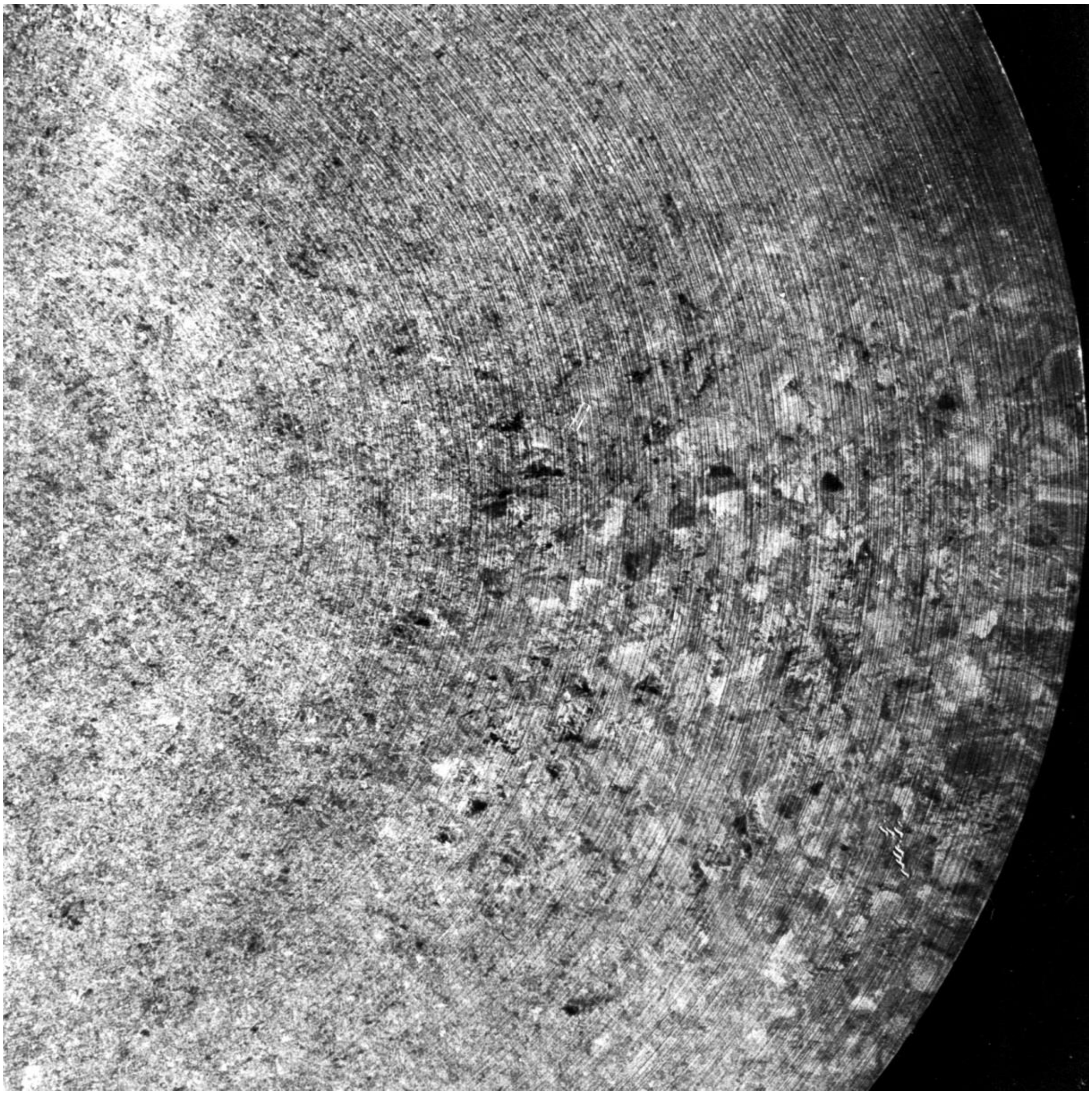
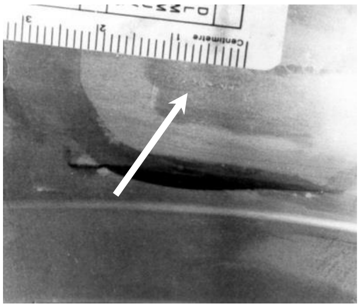


Figure 8: Example of UNACCEPTABLE macrostructure with areas of coarse grain where there is a sharply defined boundary between fine and coarse grain areas.

NEG No: 86.1145.3. (Copy Neg: 93.0855.5)



NEG No: 86.1145.3. (Copy Neg: 93.0855.5)

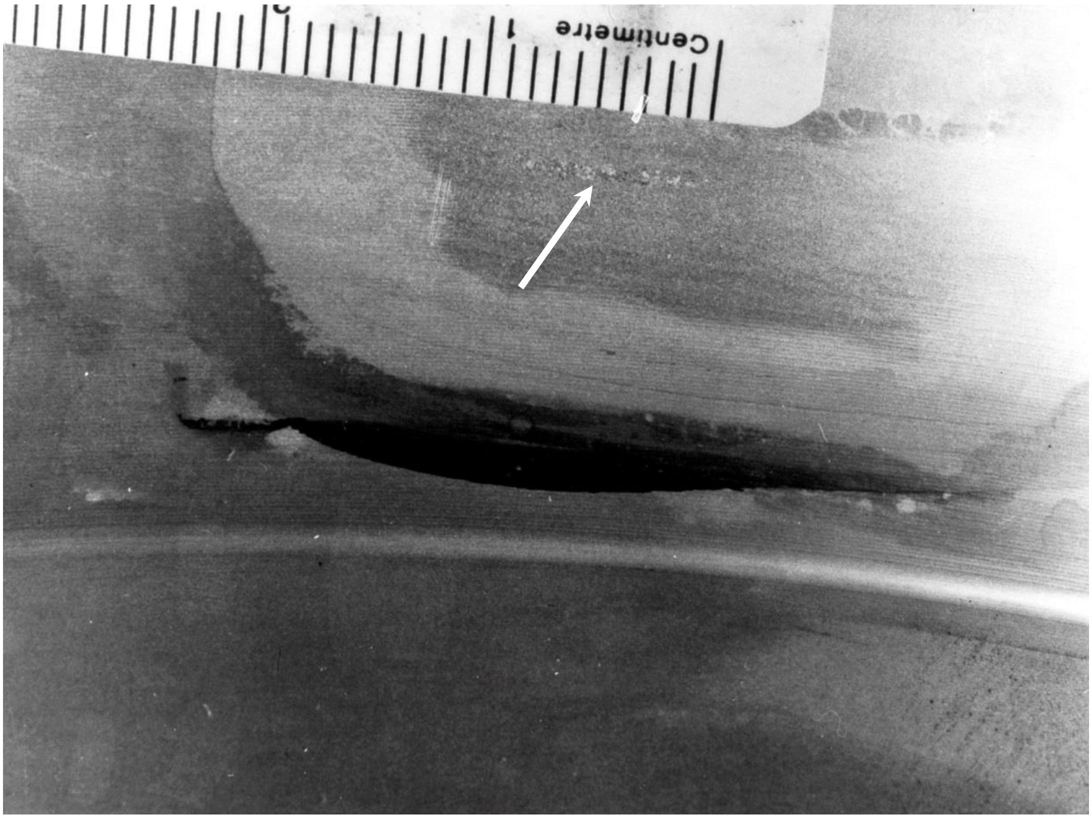


Figure 9: Example of UNACCEPTABLE macrostructure with beta fleck (top image - Mag. x1, bottom image - Mag x3.)

NEG No: H15839 (Copy Neg: 85.1084.13)

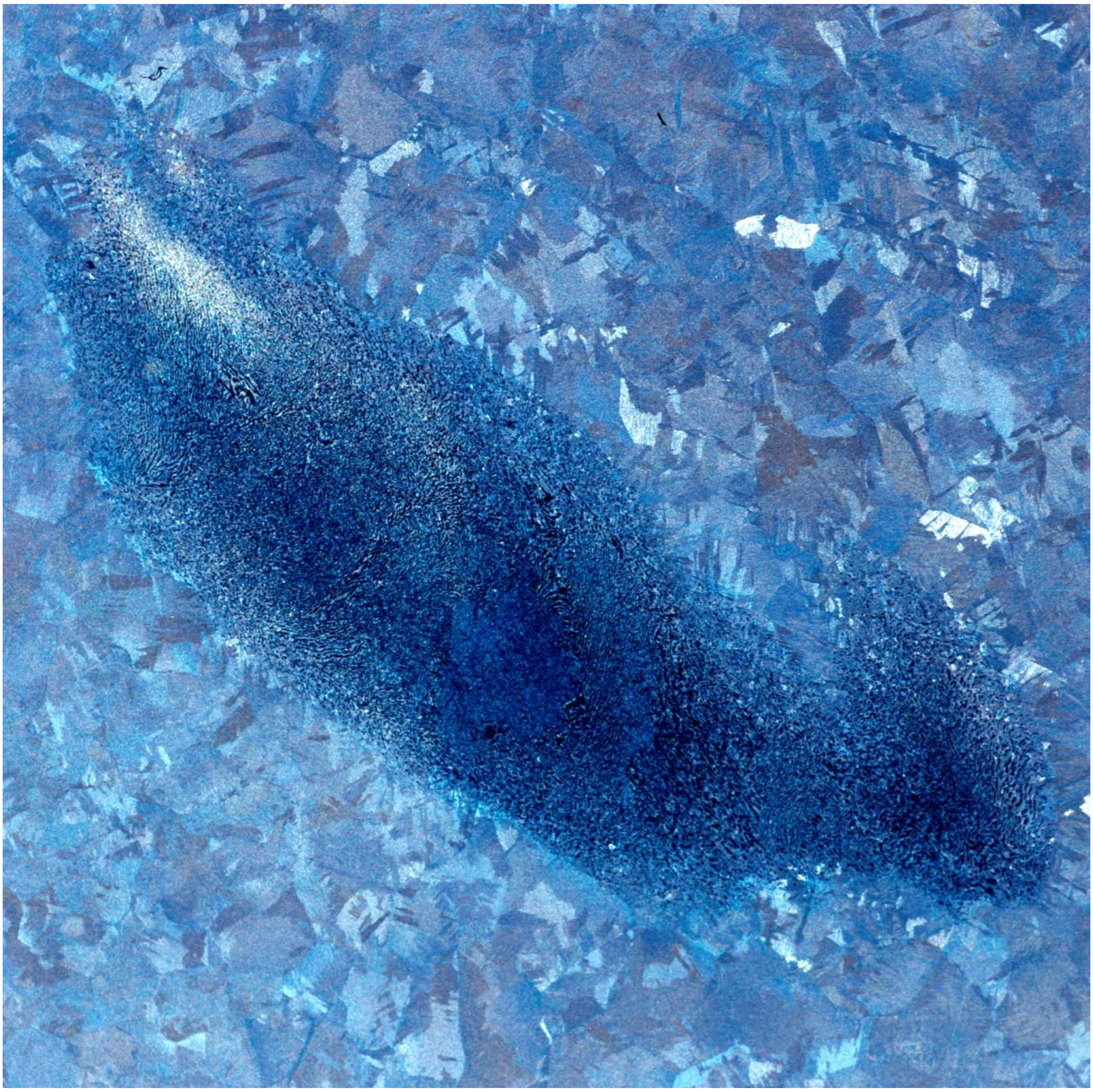
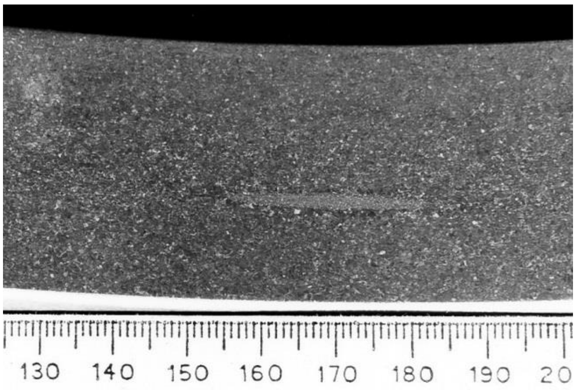


Figure 10: Example of UNACCEPTABLE macrostructure with High Interstitial Defect (HID) (Mag. x8.25).

NEG No: 88.0672.1. (Copy Neg: 93.2070.10)



NEG No: 88.0672.1. (Copy Neg: 93.2070.10)

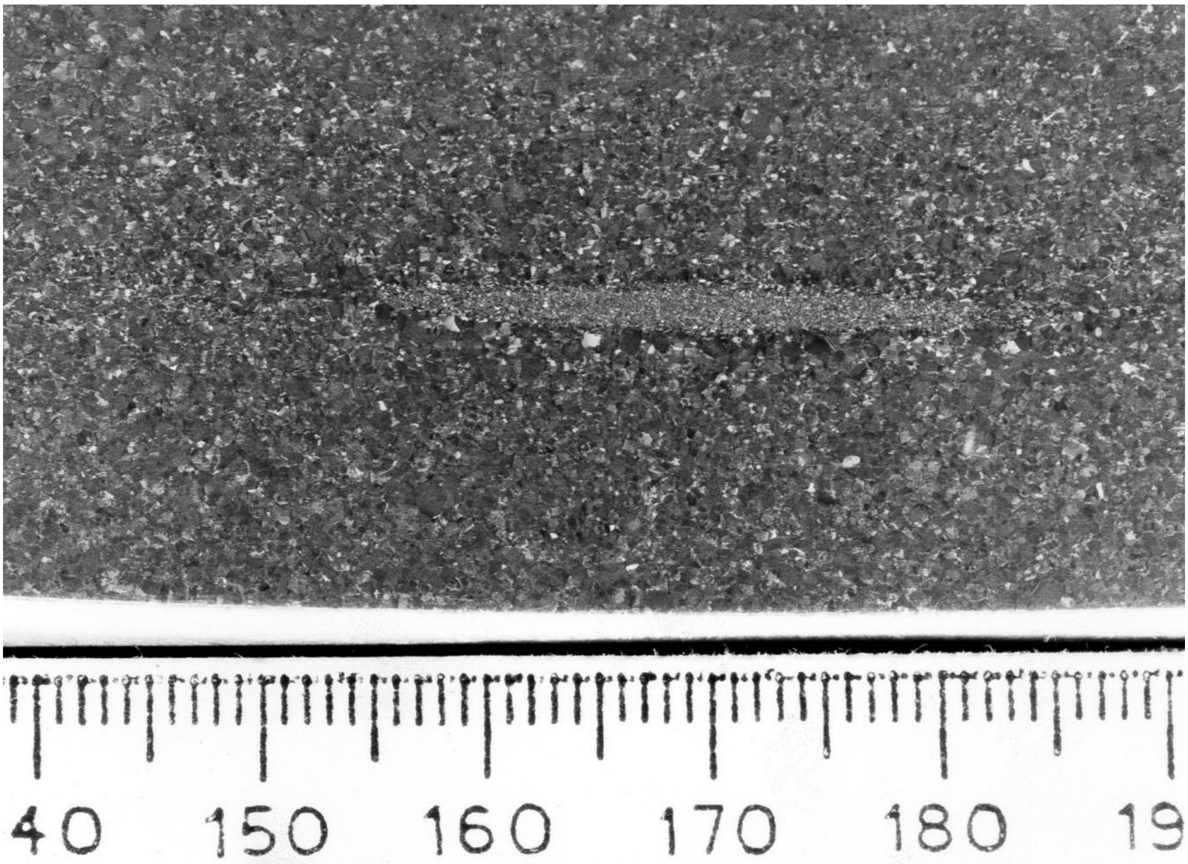


Figure 11: Example of UNACCEPTABLE macrostructure with High Interstitial Defect (HID) (top image - Mag. x1, bottom image - Mag x3.).

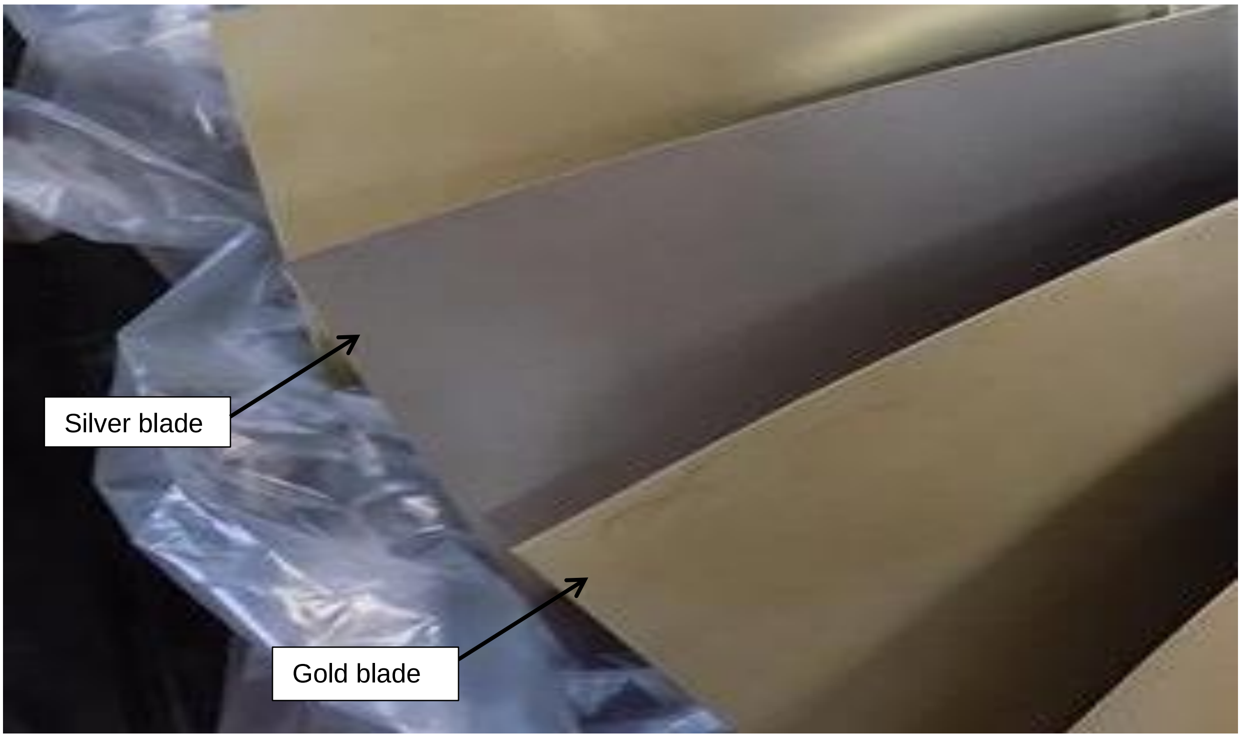


Figure 12: Examples of silver and gold discoloured Titanium 6/4 blades post heat treatment.

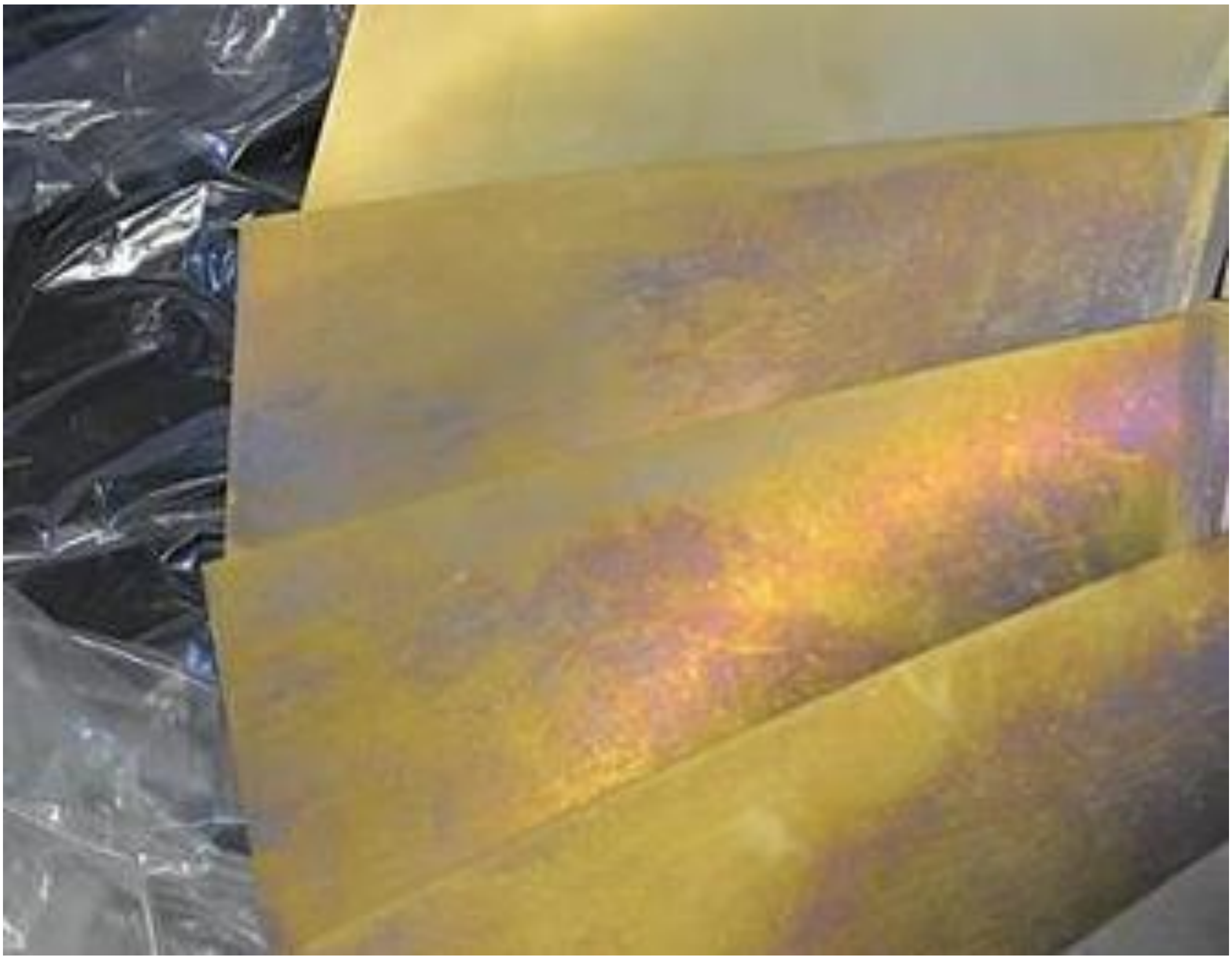


Figure 13: Examples of blue, purple and gold discoloured Titanium 6/4 blades post heat treatment

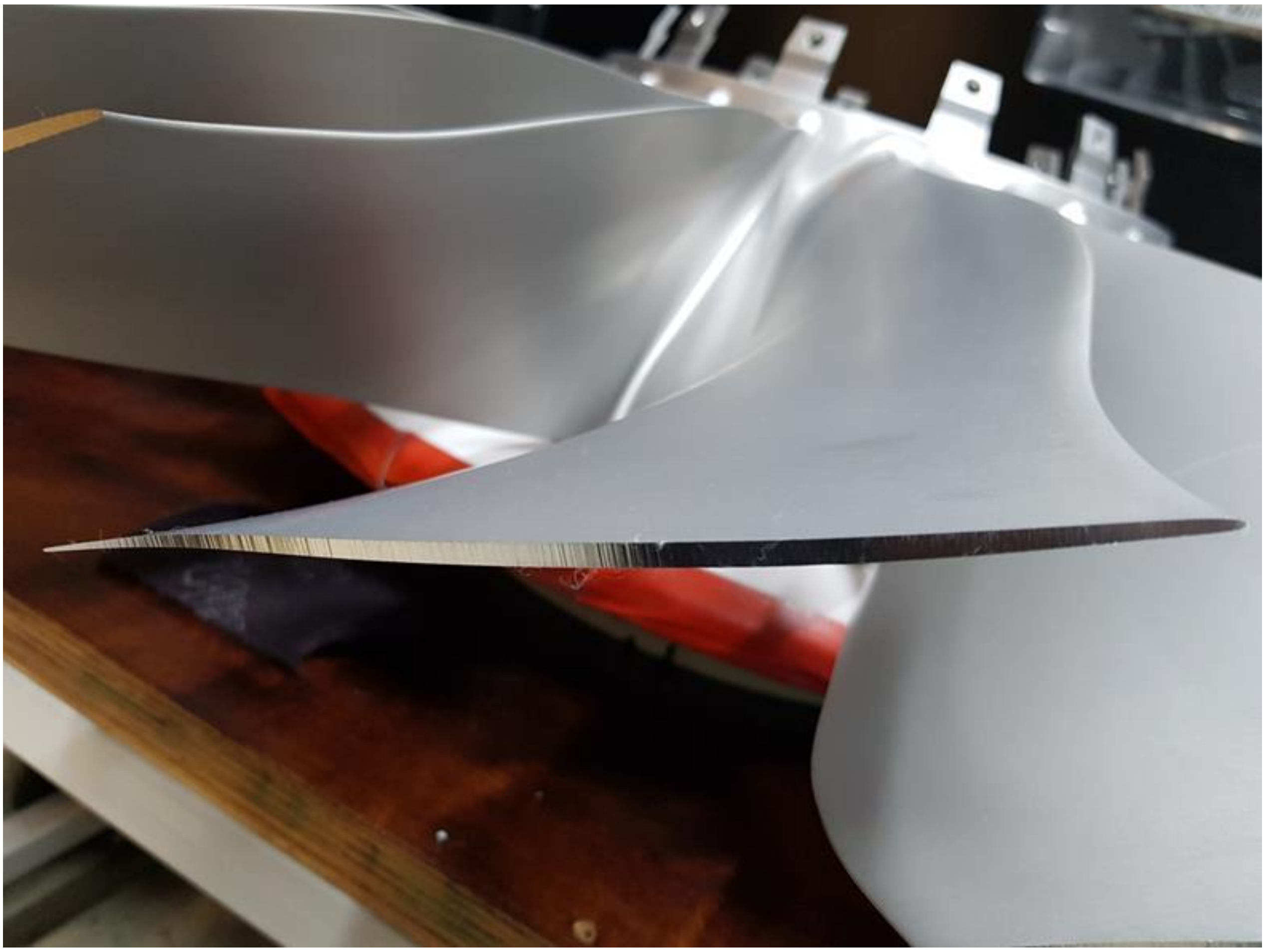


Figure 14: Tip-milled aerofoil – ACCEPTABLE machining marks, surface finish within drawing limits

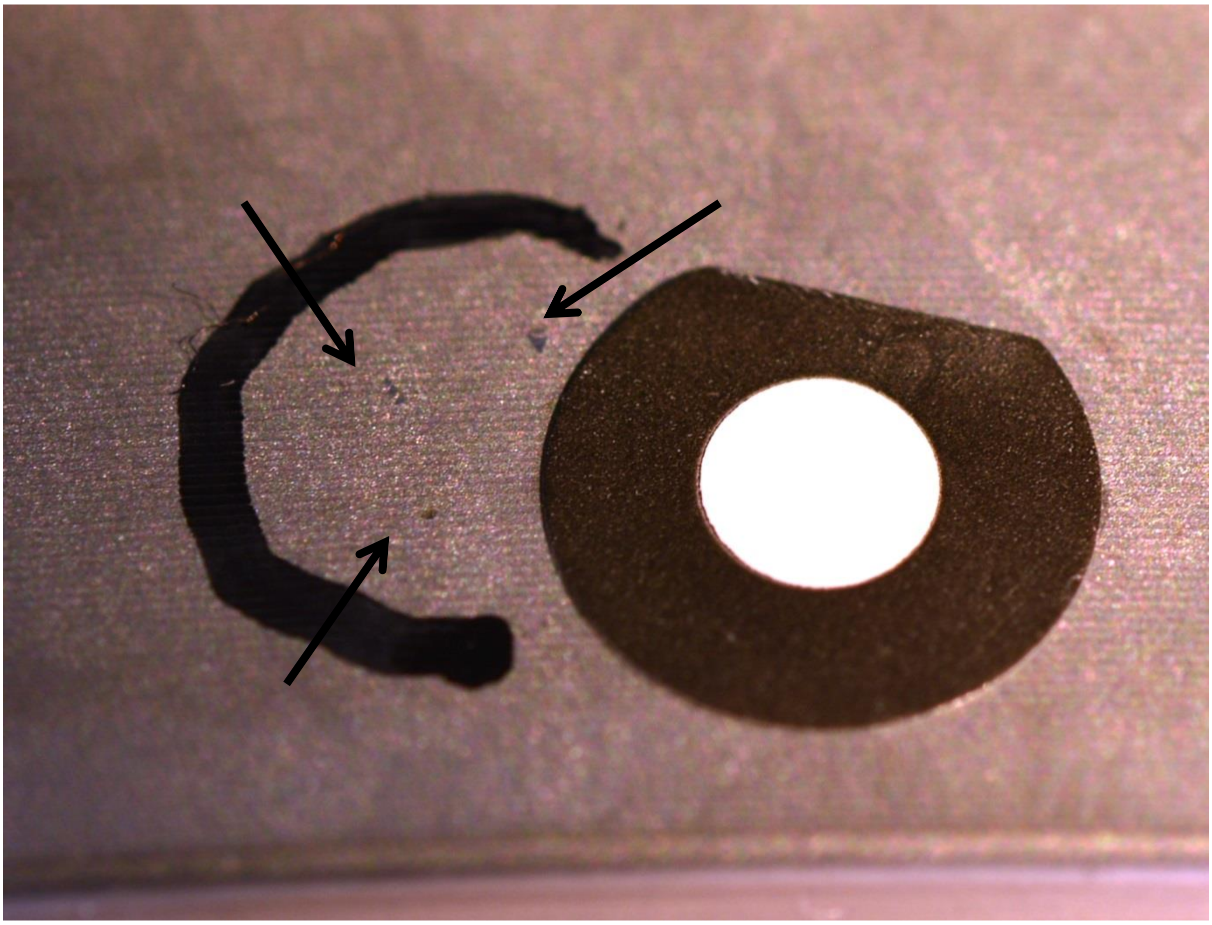


Figure 15: Example of ACCEPTABLE additional material found at etch inspection.
Marker is a 3mm white on 7mm black outside diameter.

Depth, mm	Zones (except QQ)[#]	Re-Work Requirement	Post Re-work Inspection requirements
<0,013*	All	Accept as is or lightly polish by hand with Scotch-brite or 320 grit or finer abrasive consumable product.	None
0,013 to <0,025*	Low stress	Accept as is or lightly polish by hand with Scotch-brite or 320 grit or finer abrasive consumable product.	None
	All other non-peened	Lightly polish by hand with Scotch-brite or 320 grit or finer abrasive consumable product.	Visual inspection at x10 magnification minimum to ensure that artefact has been removed.
	All other peened (both pre- and post peen)	Accept as is or lightly polish by hand with Scotch-brite or 320 grit or finer abrasive consumable product.	None
0,025 to <0,060*	Low stress	Dress by hand with Scotch-brite or 320 grit or finer abrasive consumable product.	Visual inspection at x10 magnification minimum to ensure that artefact has been removed.
	All other	Dress by hand or in line with the principles of RPS 751	Refer to RQST 2T section 3.
0,060 & deeper*	All	Dress by hand or in line with the principles of RPS 751	Refer to RQST 2T section 3.

GREY = Minor surface anomalies (scratches, scores, nicks and indents); WHITE = Major surface anomalies.

Low stress zones are given in **Section 11**.

* Verification of the depth of the feature may be carried out following removal i.e. if the depth of polishing to remove the anomaly is <0,013mm, it follows that the depth of the anomaly was <0,013mm.

[#] For rework of feature type QQ refer to section 5 of this RQSC. For acceptance of surface anomalies in area QQ refer to section 5.4.9 of this RQSC.

Figure 16: Summary of Re-Work and Inspection Requirements for Surface Anomalies
Note: Surface anomalies are scratches, scores, nicks and indents.

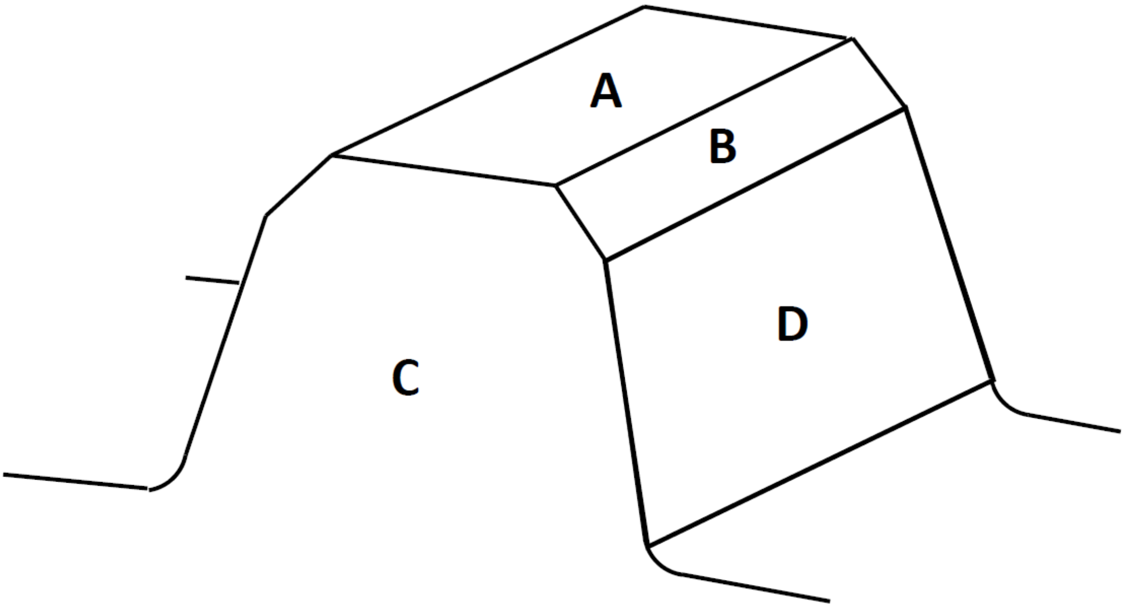


Figure 17: Curvic Tooth Zoning Diagram

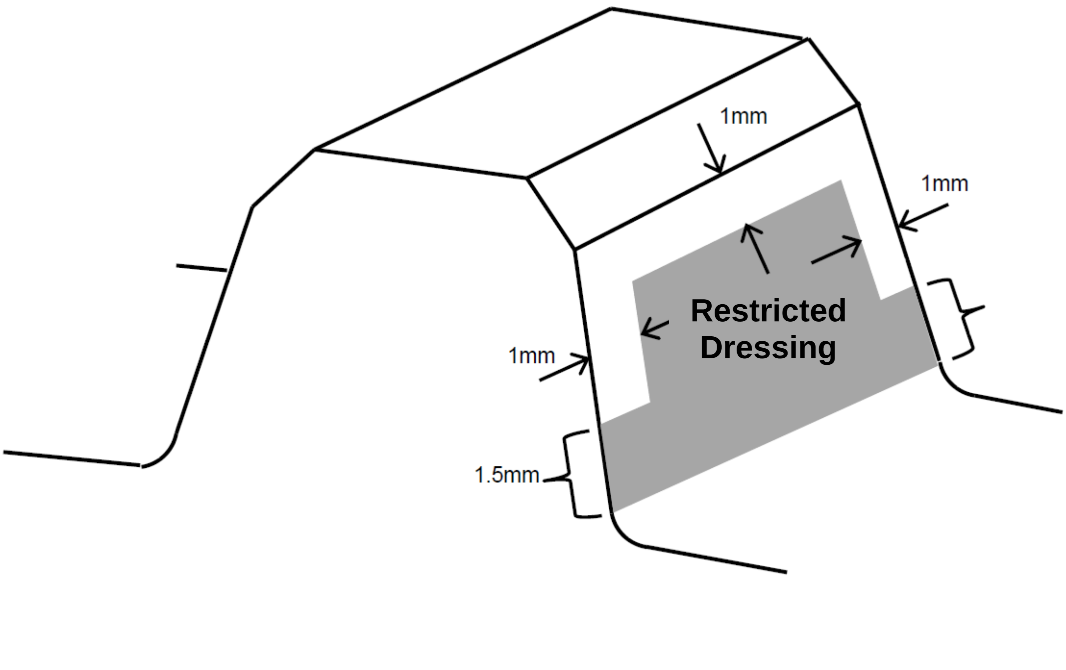
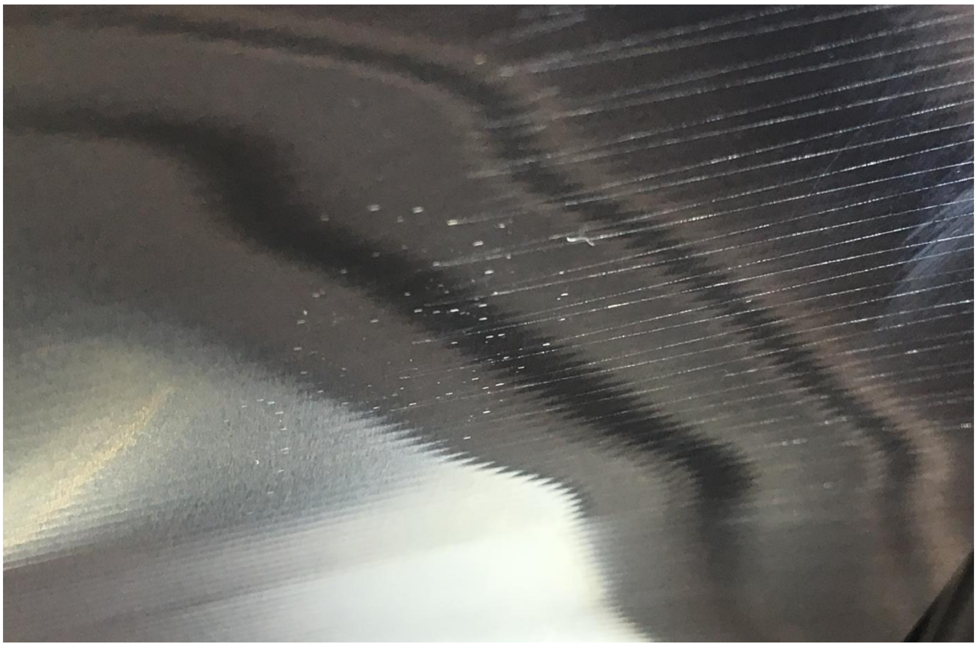
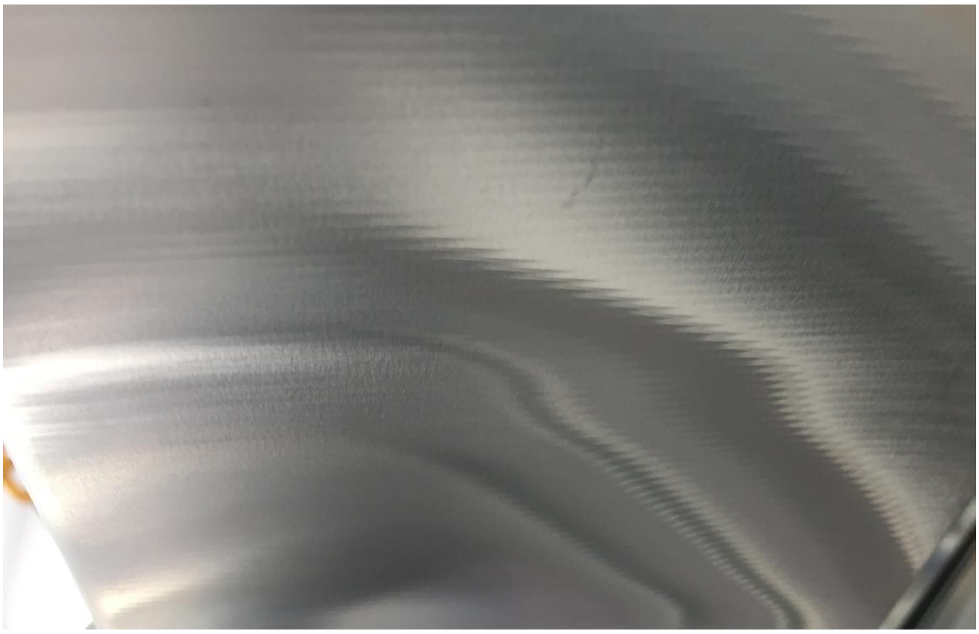


Figure 18: Curvic Tooth Dressing Restriction

Figure 19: Example of ACCEPTABLE visual indications with no discernible depth.



a) - Example of ACCEPTABLE visual indications on aerofoil with no discernible depth (image taken under directed white light)



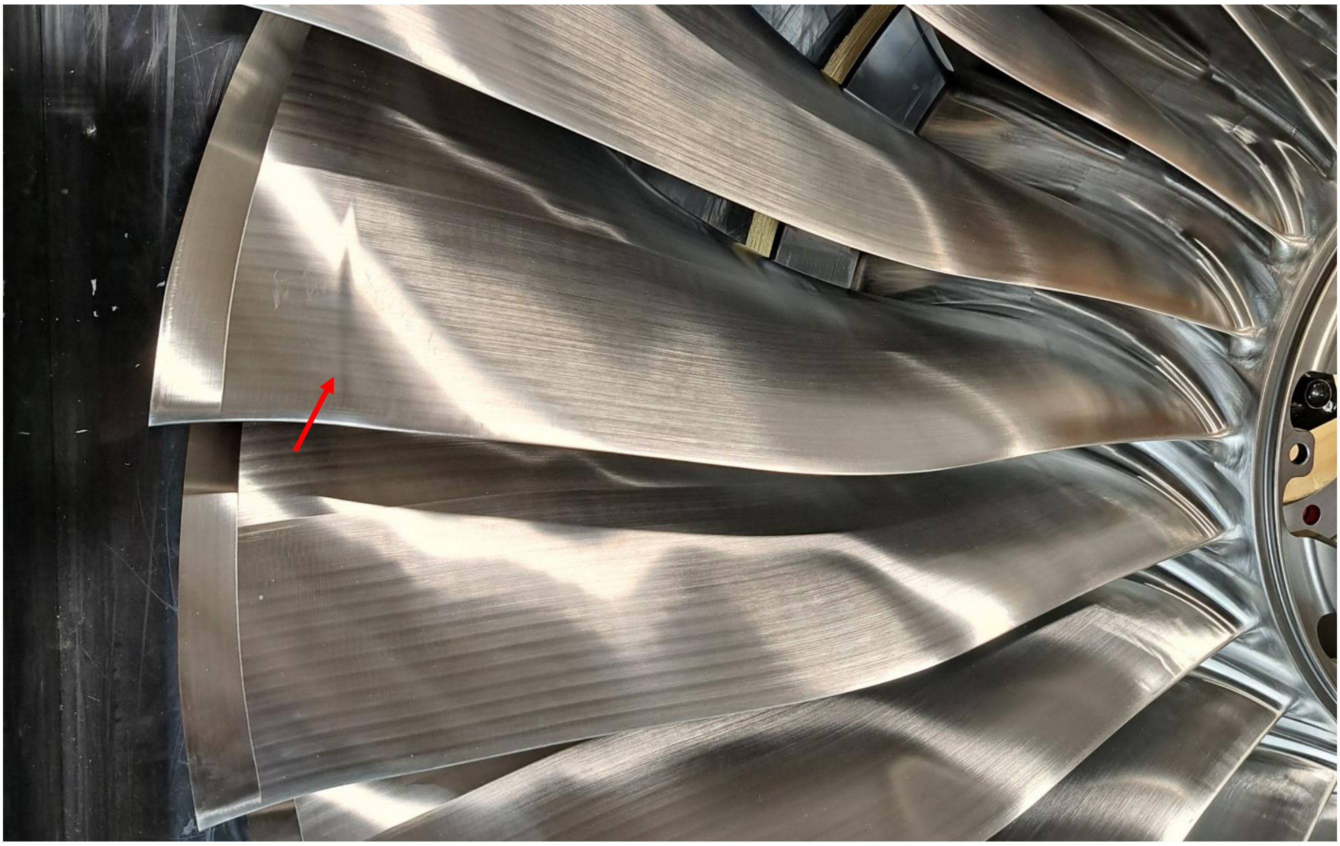
b) – Same aerofoil and part as shown in Figure 20a, in normal light – ACCEPTABLE

Figure 20: ACCEPTABLE visual indications

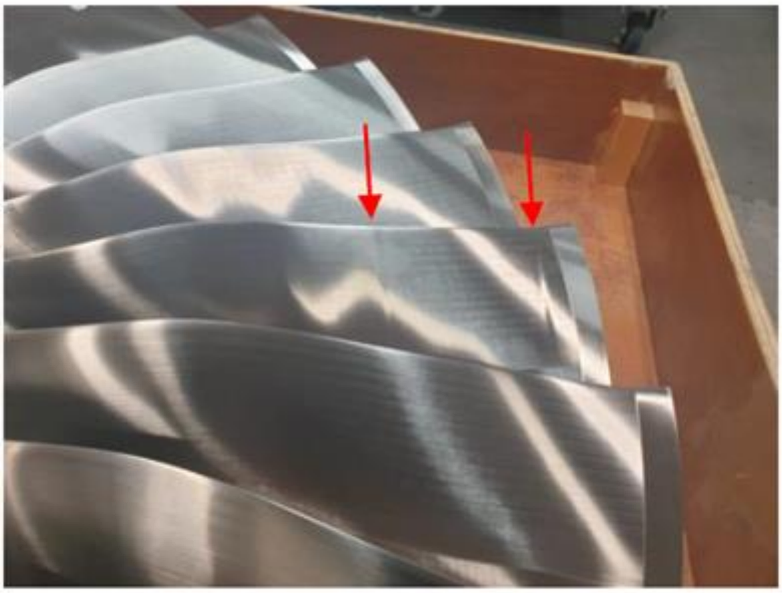
a) – Example of ACCEPTABLE visual machining marks on a blade.

b) – Example of ACCEPTABLE visual machining marks during blisk manufacture.

Figure 21: ACCEPTABLE machining marks on aerofoils



a) – Example of ACCEPTABLE undulations



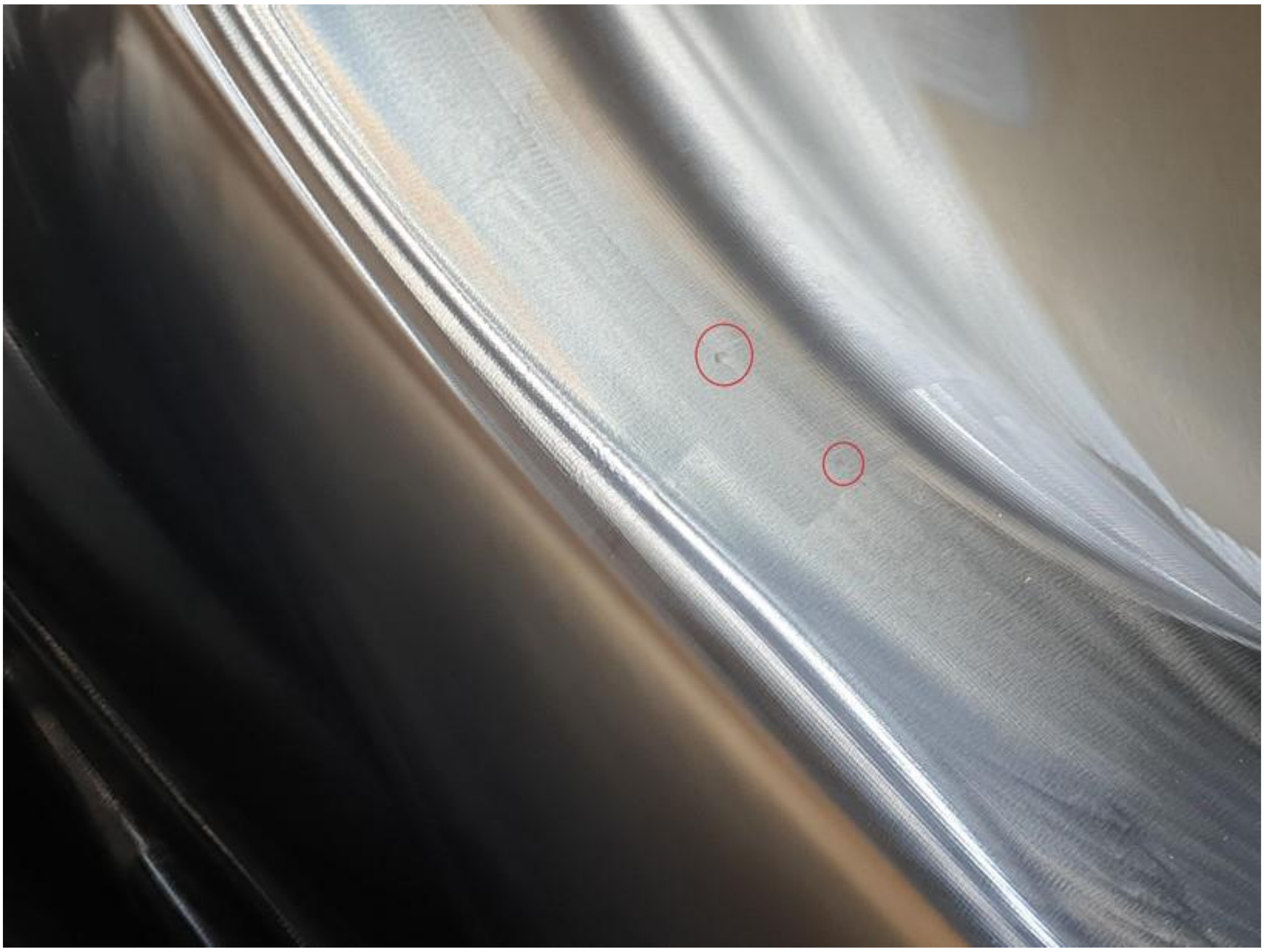
b) – Example of ACCEPTABLE mismatches

Figure 22: ACCEPTABLE undulations and mismatches
Note: These are images are examples only, and not a complete list

a) – Example of ACCEPTABLE machining marks

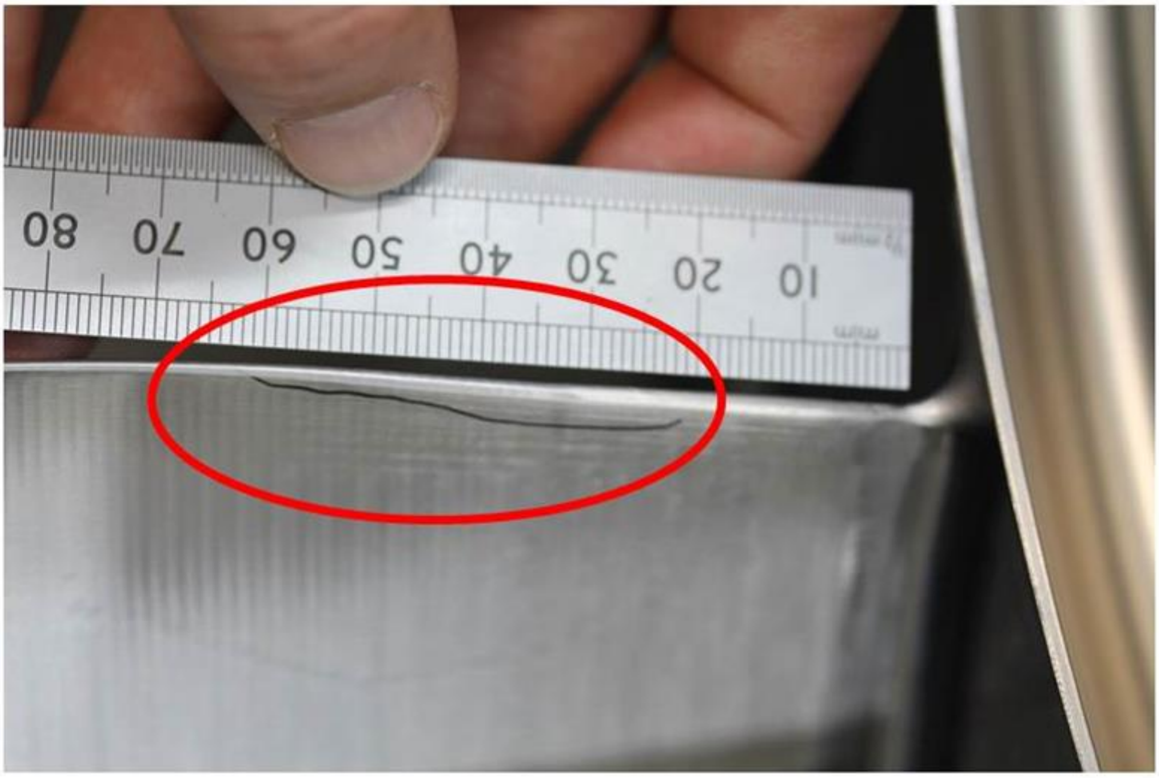
Figure 23: ACCEPTABLE examples of machining marks on polished aerofoil leading and trailing edges, and dwell marks in the fillet and annulus
Note: These are images are examples only, and not a complete list

Continued Overleaf

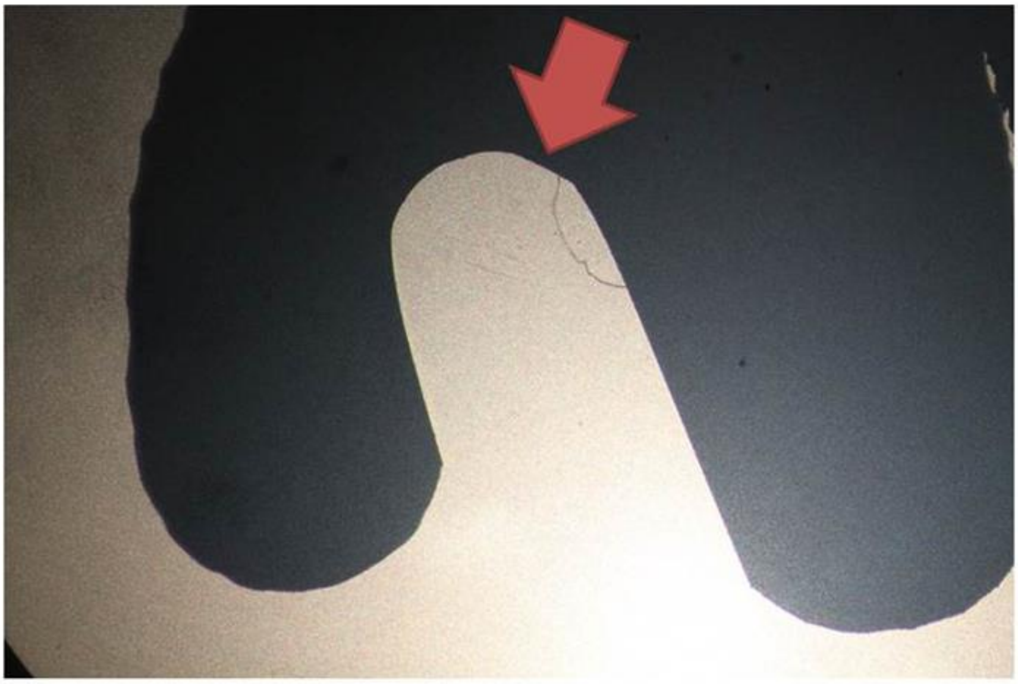


b) – Example of ACCEPTABLE dwell marks

Figure 23: ACCEPTABLE examples of machining marks on polished aerofoil leading and trailing edges, and dwell marks in the fillet and annulus
Note: These are images are examples only, and not a complete list



(a) Example of ACCEPTABLE flats on aerofoil trailing edge.



(b) Example of ACCEPTABLE flat on aerofoil TE (shadowgraph image).

Figure 24: ACCEPTABLE flats on trailing edges of aerofoils

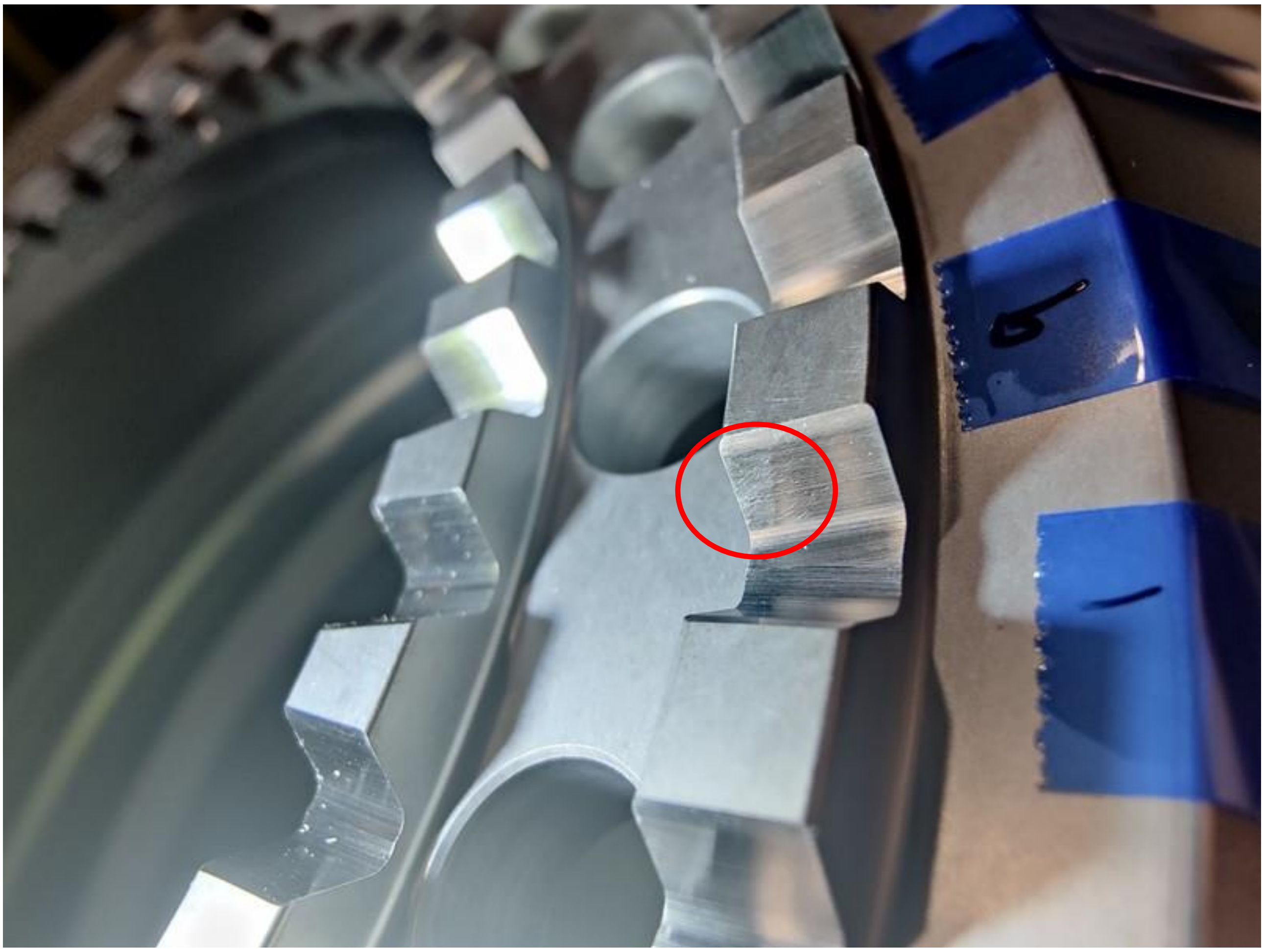


Figure 25: ACCEPTABLE visual appearance of curvic teeth.
Marks are from production deburring of the teeth edges.

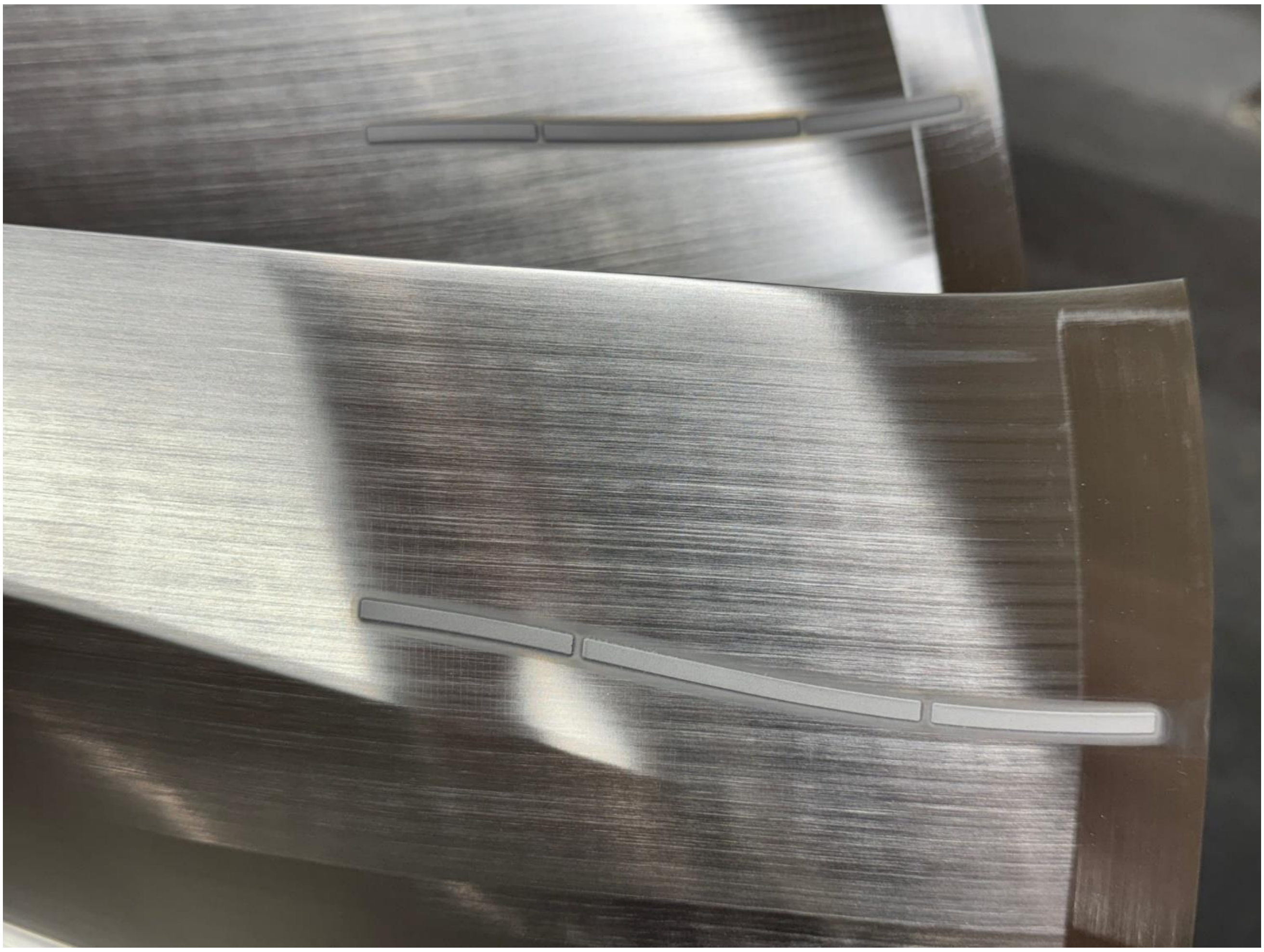


Figure 26: ACCEPTABLE visual appearance of transition (turbulator) strips. **(continued overleaf)**

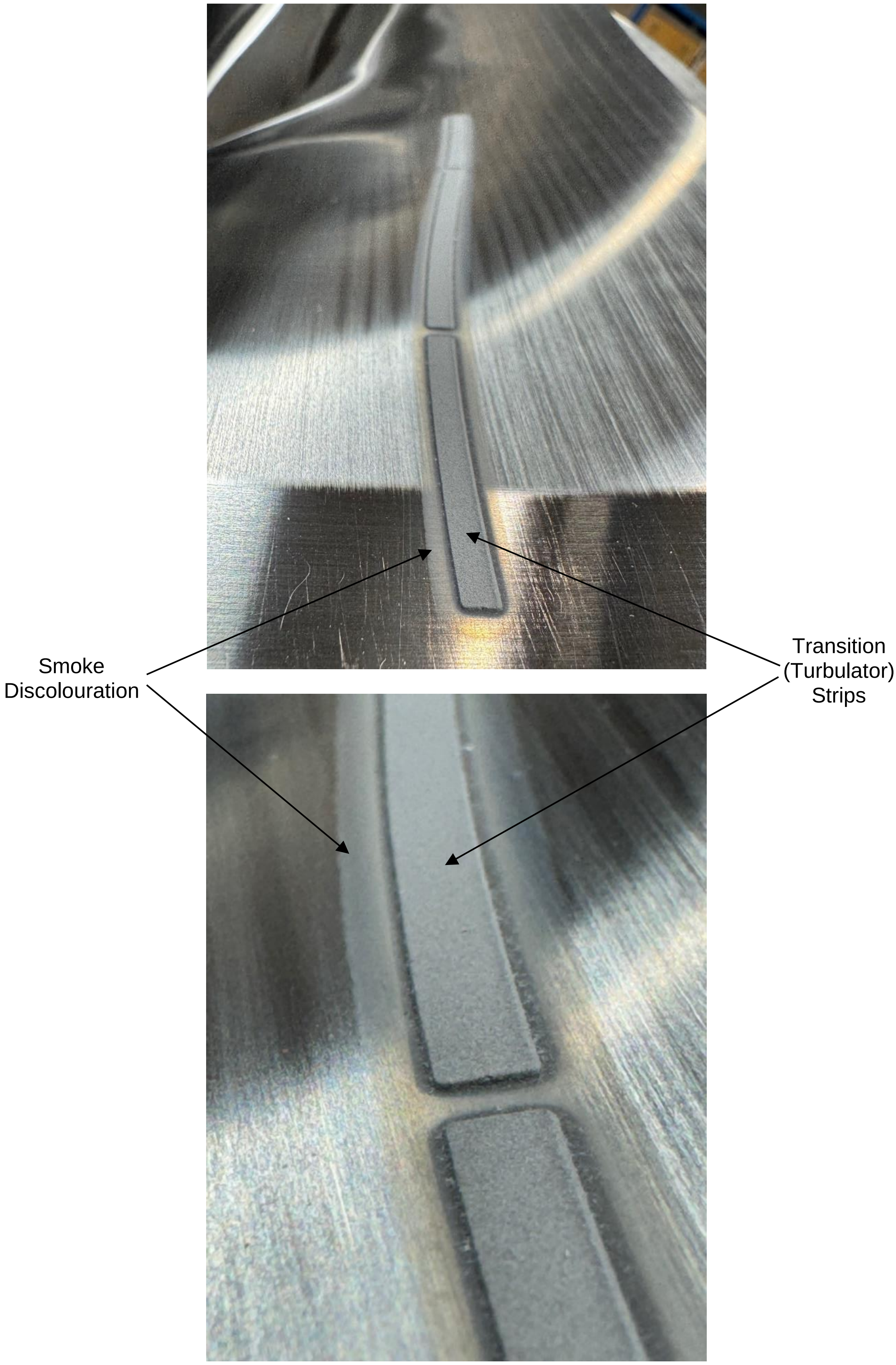


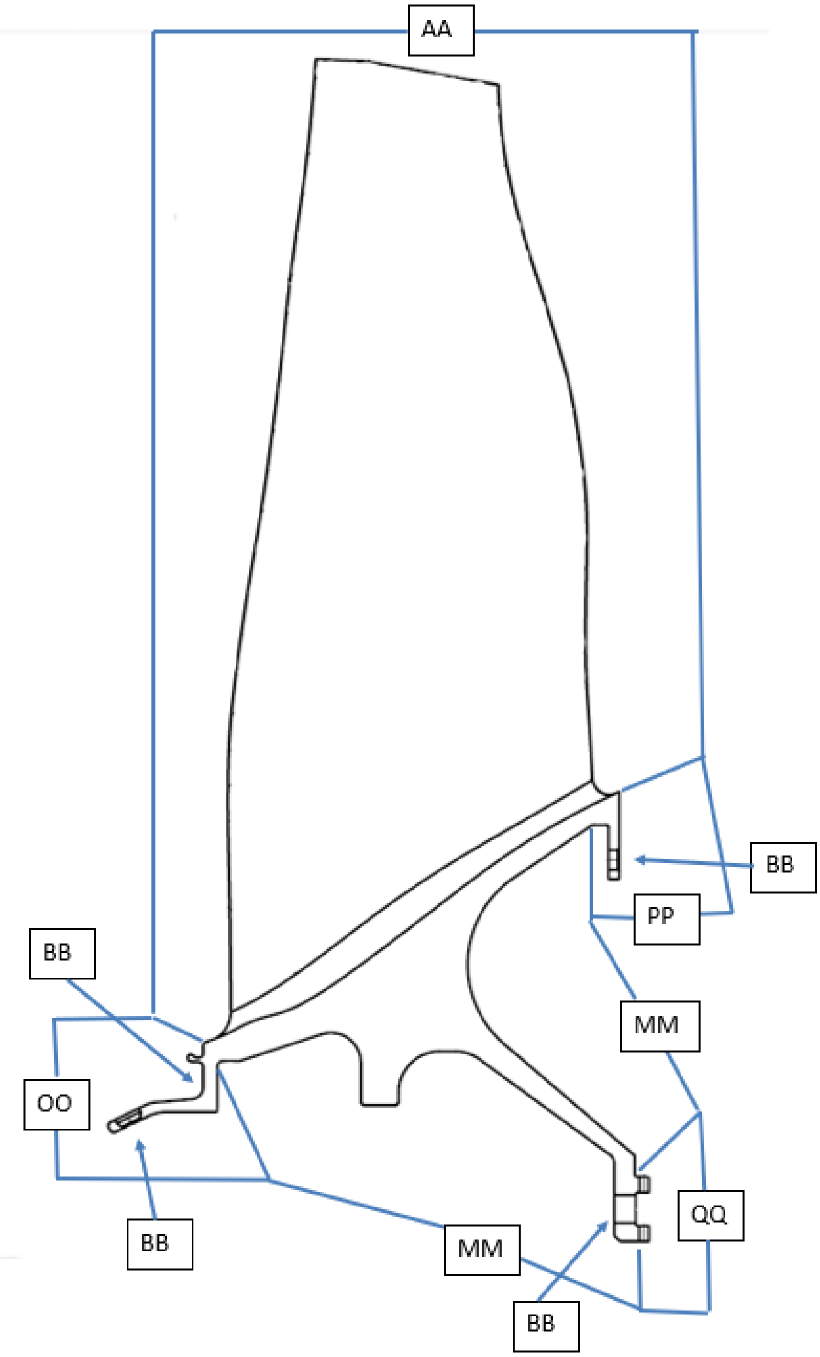
Figure 26: ACCEPTABLE visual appearance of transition (turbulator) strips.

11.0 ZONE DIAGRAMS

Note: Where required part number is not given then Section 11.1 is to be used.

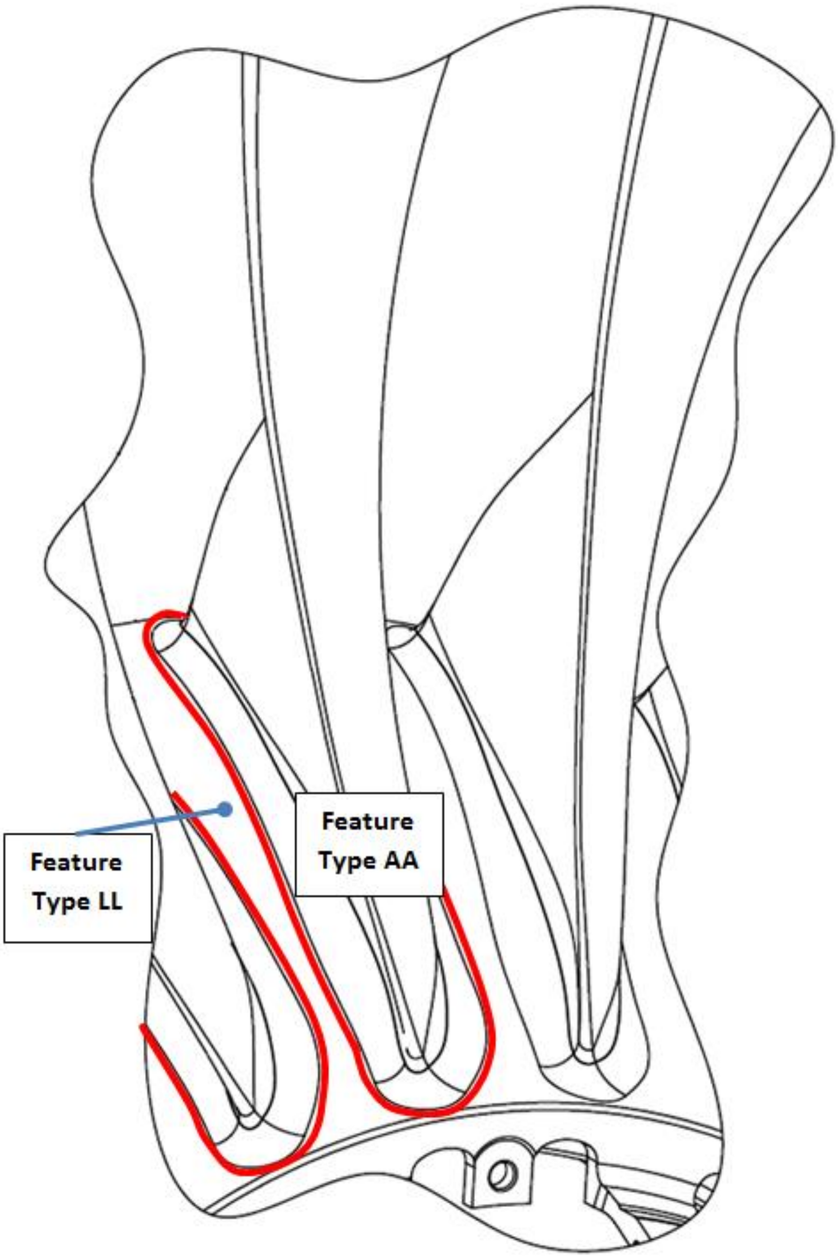
11.1 Zone Diagrams for production standard part numbers LV25569, LV39925 and LV30633

11.1.1 Blisk Feature Zones (see 11.1.2 for feature type LL) - LV25569, LV39925 and LV30633

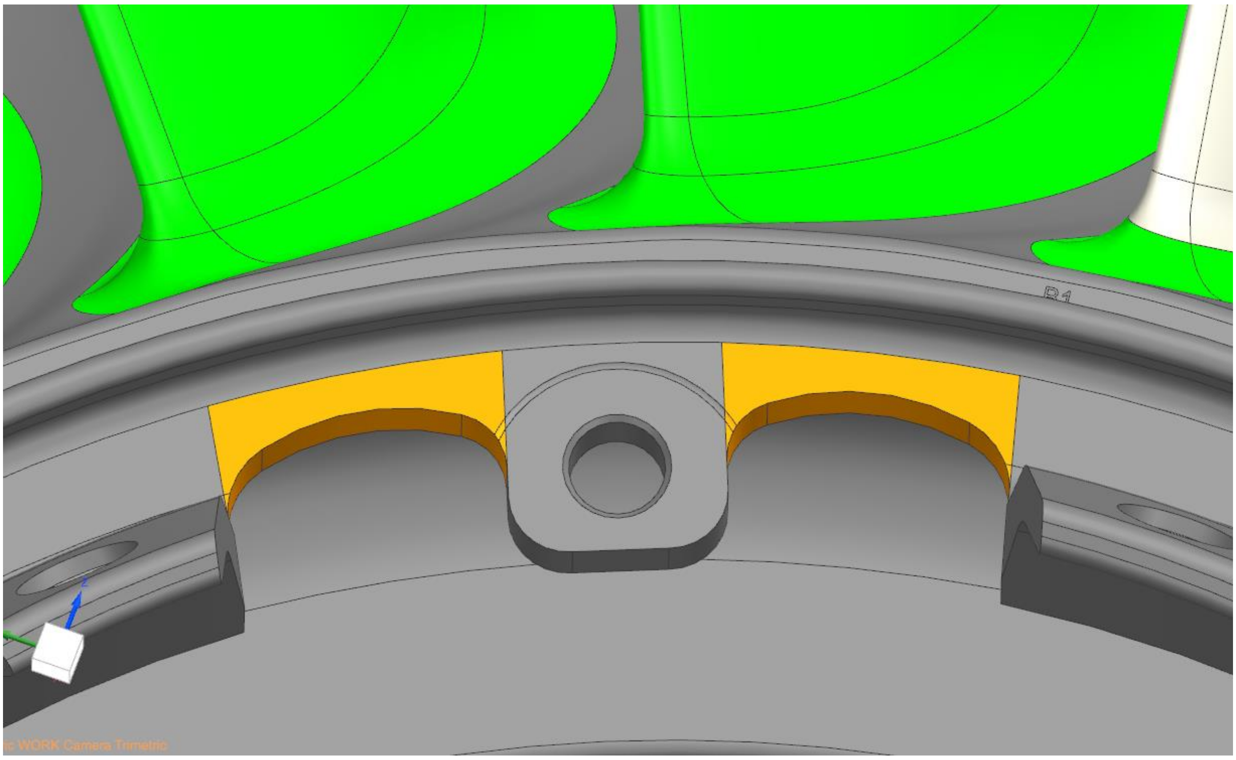


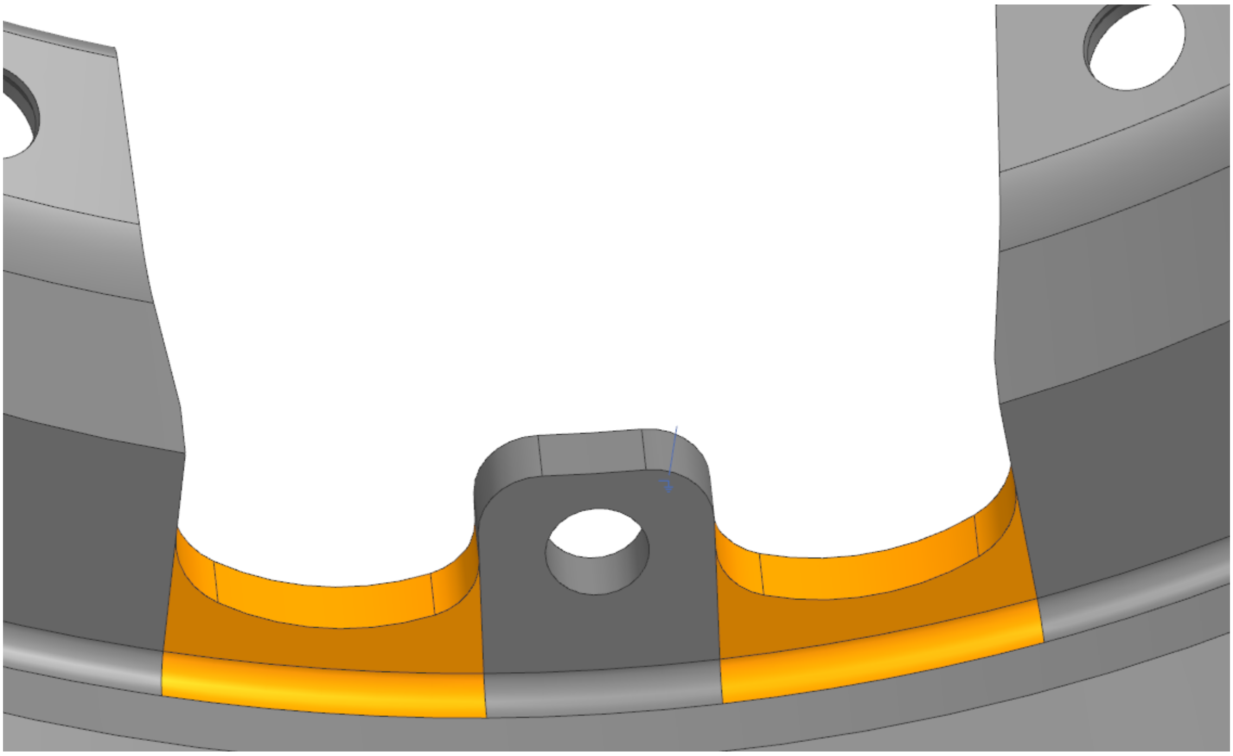
11.1.2 Fillet and Annulus Transition - LV25569, LV39925 and LV30633

Note: Transition from feature type AA to LL is 3mm on from where blade fillet runs into the annulus

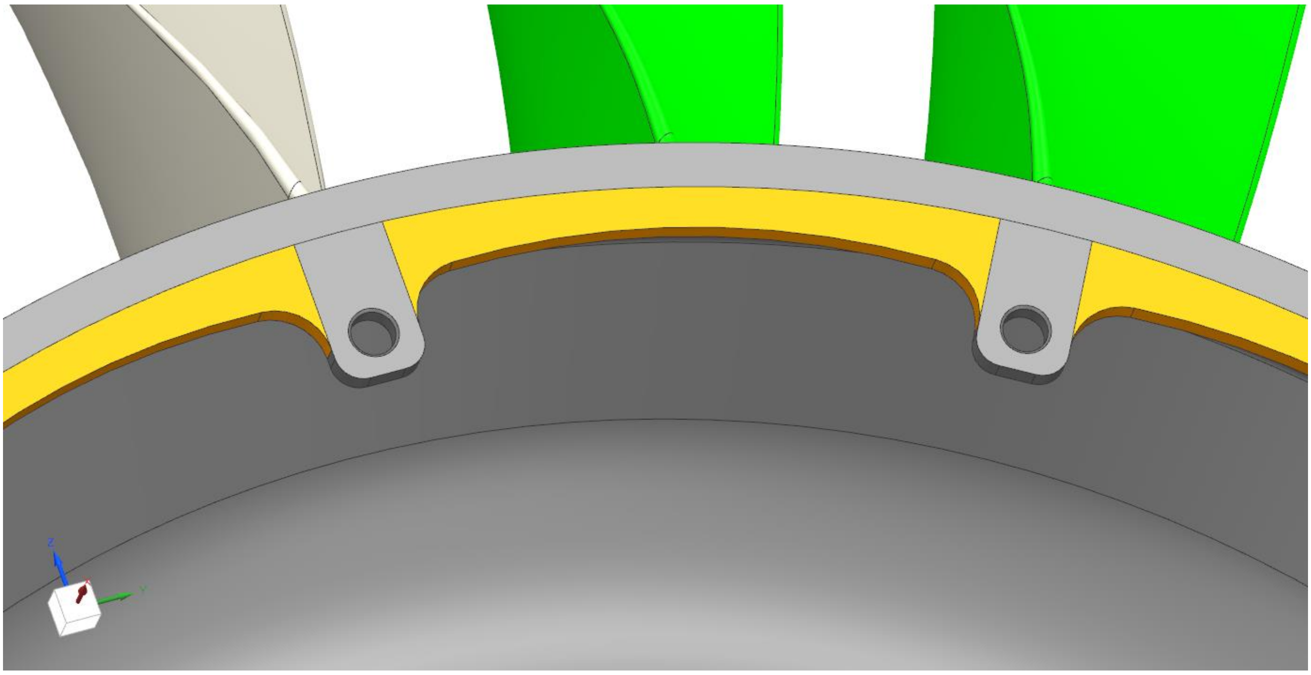


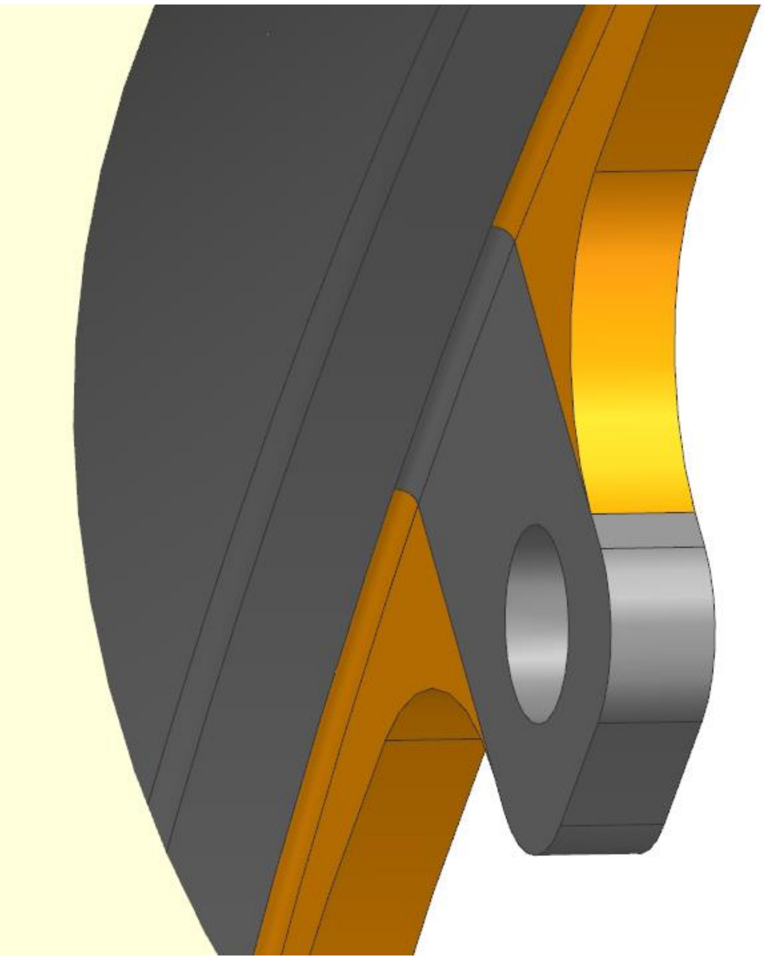
11.1.3 Low Stress Zones - LV25569, LV39925 and LV30633





(a) **All areas of feature type OO are low stress except the scallop radii, as shown in 'Orange' above**
Note: Holes are feature type BB.

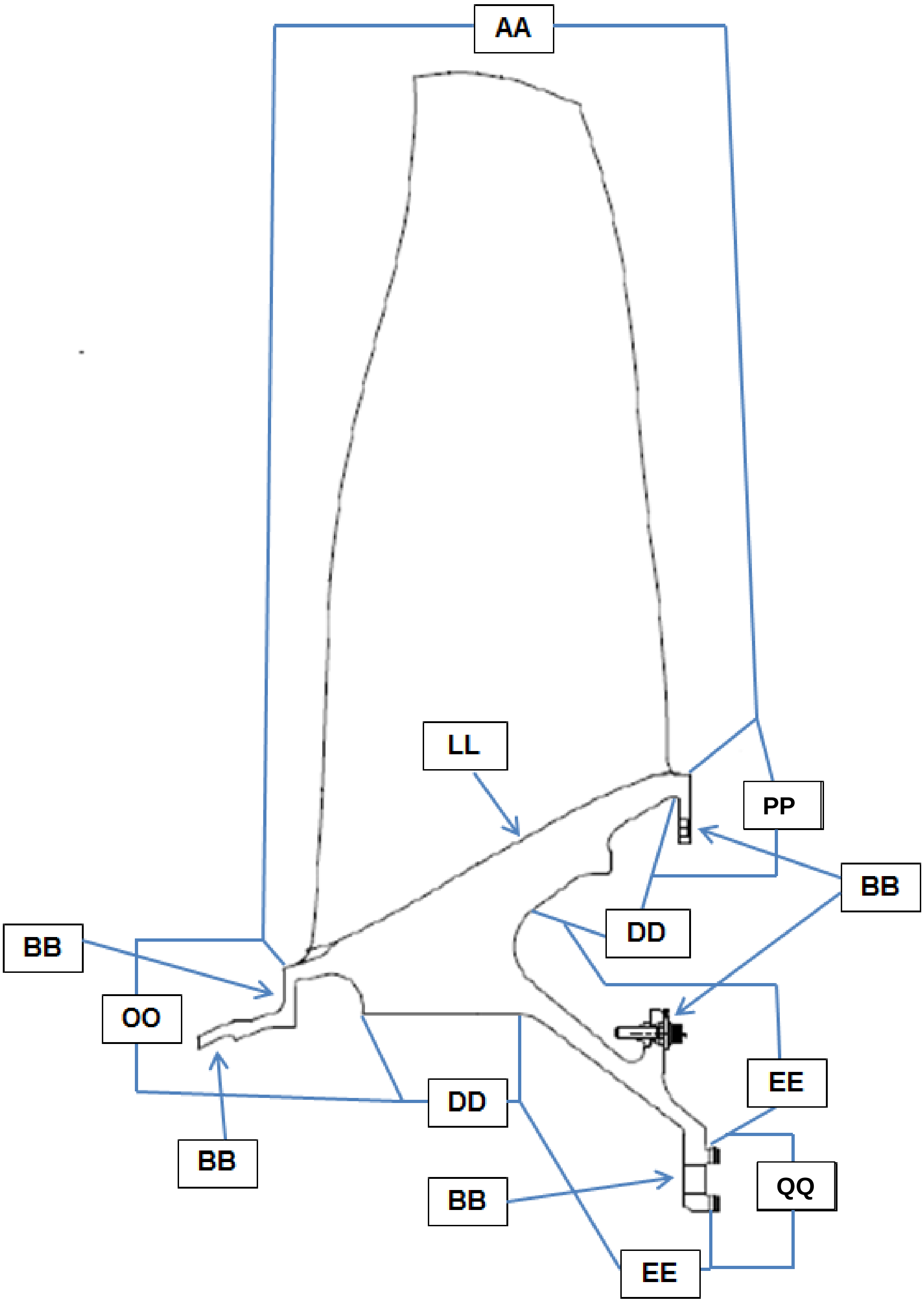




(b) All areas of feature type PP are low stress except the trailing edge scallop and radii, as shown in 'Orange' above
Note: Holes are feature type BB.

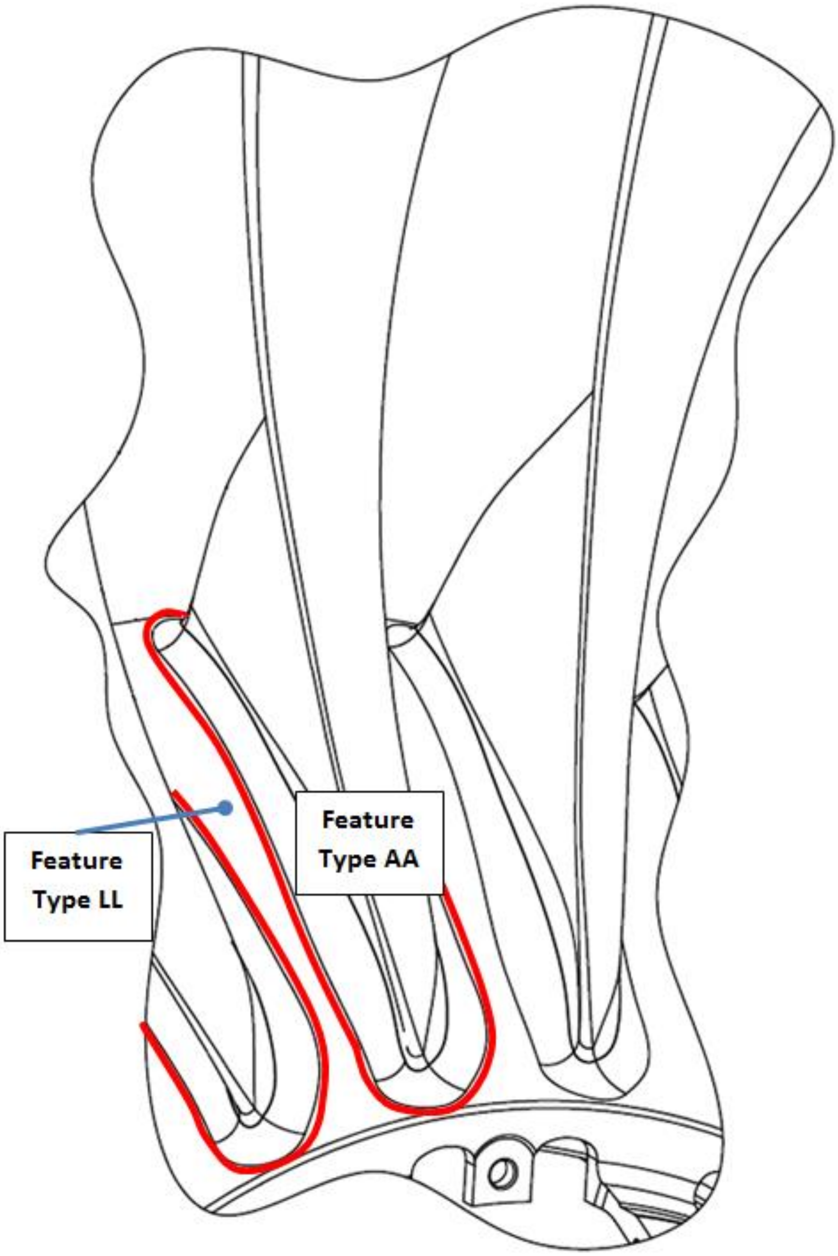
11.2 Zone Diagrams for production standard part numbers KH55017, LV10131 and LV10002

11.2.1 Blisk Feature Zones (see 11.2.2 for feature type LL) - KH55017, LV10131 and LV10002

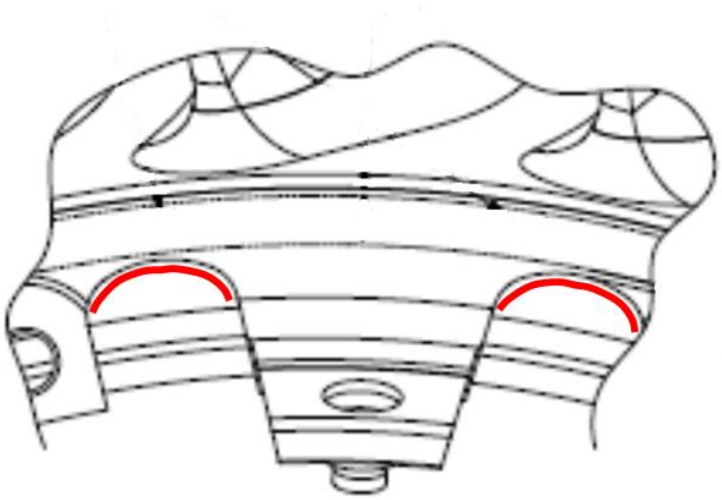


11.2.2 Fillet and Annulus Transition- KH55017, LV10131 and LV10002

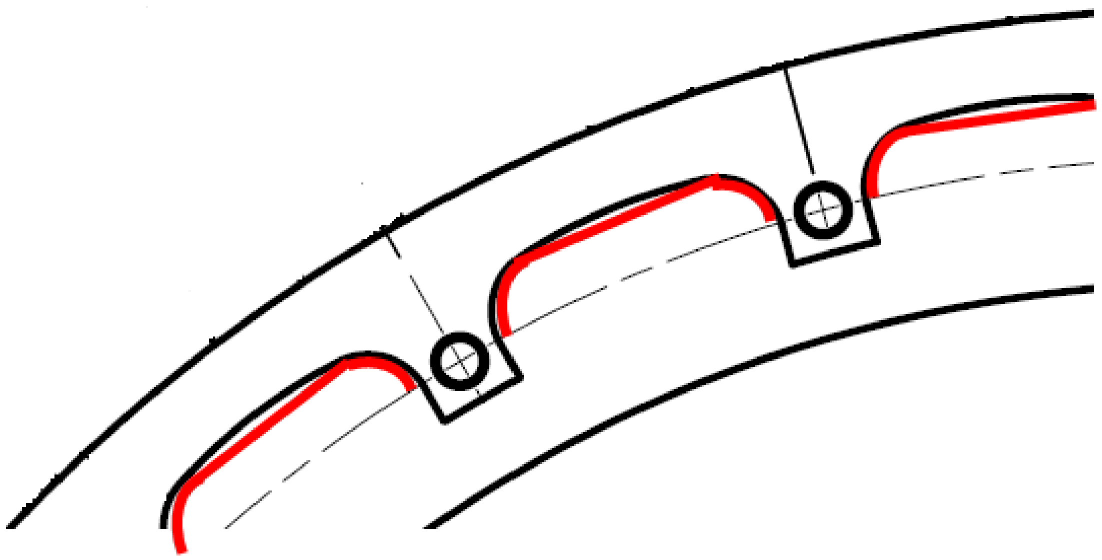
Note: Transition from feature type AA to LL is 3mm on from where blade fillet runs into the annulus



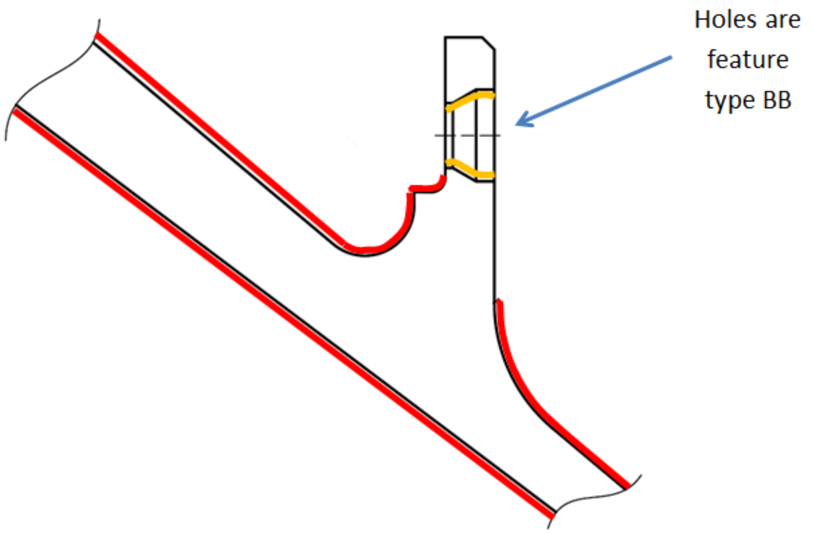
11.2.3 Low Stress Zones- KH55017, LV10131 and LV10002



(a) All areas of feature type OO are low stress except the scallop radii, as shown in ‘red’ above
Note: Holes are feature type BB.



(b) All areas of feature type PP are low stress except the trailing edge scallop and radii, as shown in ‘red’ above
Note: Holes are feature type BB.



(c) **Damper flange of feature type EE is low stress. This is only region shown above that is not coloured. Other areas of feature type EE are shown in 'red'**
Note: Damper flange holes are feature type BB and are shown in 'orange' above with nut, bolt and washer removed.

12.0 REASON FOR ISSUE

12.1 This standard contains the following changes from the previous standard. The report(s) referenced confirms that the standard, or changes to the standard, have no impact on product integrity.

Page(s)	Change Summary	Report Reference
All	Change of format to latest RQSC format from when re-written in 2022 / 23. Correction of typographical errors, along wording changes to clarify and simplify requirements.	CRFLR 2022/29
3	In scope, statement regarding issue 2 removed and statement regarding components manufactured prior to issue 3 added.	CRFLR 2022/29
4, 7, 13 &14	Change to acceptance for beta ghost grains, and to acceptance of images in figures 2&3	CRFLR 2022/29
5, 6 & 7	Sections on post heat treatment visual inspection and post heat treatment discolouration acceptance have been combined. New sections for heat treatment to allow acceptance of cycle aborts, load thermocouple failures and partial loss of vacuum. Section for cycle assessment added. New clauses allowing acceptance of machining marks from blades, flats on aerofoil trailing edges, deburr marks on curvic teeth and score marks in bolt holes after shank nuts / balance bolts have been removed.	CRFLR 2022/29
8, 12, 37 & 38	Transition (turbulator) strips accept / reject criteria added and rework allowance along with Figure 26 added.	CRFLR 2022/29
9, 10 & 11	Rework section items added to clarify instructions. Feature type MM (under rim) added. Repeat statement in feature type BB removed. Feature type QQ completely re-written.	CRFLR 2022/29
39-46	Figures containing Blisk feature zones, fillet and annulus transition and low stress zones moved for Appendices to section 11. Other figures re-numbered. Also new set of images added for current design of RB3043 / RB3070 components.	CRFLR 2022/29
12	Additional definitions added covering non-peened surfaces, load thermocouple failures and partial loss of vacuum.	CRFLR 2022/29
28	“Do Not Dress” area on Curvic tooth restricted dressing changed to “Restricted Dressing” area.	CRFLR 2022/29
29-38	Figures 19 and 20 of acceptable visual indications added. Figures 21 & 23 of acceptable machining marks added. Figure 22 of acceptable undulations and mismatches added. Figure 24 of acceptable flats on aerofoil trailing edges added. Figure 25 of acceptable deburr marks on Curvic teeth added. Figure 26 of transition strips added.	CRFLR 2022/29
39-46	Section 11 for zone diagrams added.	CRFLR 2022/29